

# INDIAN SPACE OLYMPIAD (ISO) 2026

**ADVANCED LEVEL - CLASSES XI-XII**

## FIVE FULL-LENGTH MOCK QUESTION PAPERS

*JEE Main-Level Difficulty | Strictly Syllabus-Aligned*

**ASTRONOMY - SPACE SCIENCE - PHYSICS - REASONING**

Practice Edition

Information cut-off for current-space questions: 23 July 2026

## Scope and Examination Design

This volume contains five independent 50-question mock papers for the ISO Advanced Level (Classes XI-XII). Every question is mapped to one of the eleven topics in the supplied Advanced Level syllabus. No question requires a topic outside that syllabus. The phrase “JEE Main-level” refers only to the intended standard of physics, mathematics, data interpretation and multi-step reasoning; these papers are not issued by or affiliated with the National Testing Agency or JEE.

### Pattern of Each Paper

| Section | Questions | Marking                         | Level                                              |
|---------|-----------|---------------------------------|----------------------------------------------------|
| A       | 1-40      | +4 correct, -1 incorrect        | Q1-Q5 warm-up; Q6-Q40 predominantly JEE Main-level |
| B       | 41-50     | +4 correct, no negative marking | Numerical-answer questions                         |
| Total   | 50        | 200 marks                       | Suggested time: 150 minutes                        |

### Strict Syllabus Coverage

- 1. Fundamentals of Astronomy: observational astronomy, celestial mechanics, coordinates, spectroscopy, photometry and instruments.
- 2. The Solar System: planetary science, atmospheres, orbital mechanics, geology, small bodies and exoplanets.
- 3. Stars, Galaxies & the Universe: stellar evolution and remnants, galaxies, introductory cosmology, Big Bang, dark matter and dark energy.
- 4. Earth & Moon: Earth sciences, atmosphere, planetary environments, lunar evolution and habitability.
- 5. Satellites & Space Technology: engineering, spacecraft systems, attitude control, communications, Earth observation and mission operations.
- 6. Rockets & Launch Vehicles: propulsion, launch systems, orbital mechanics, cryogenic propulsion, planning and reusability.
- 7. ISRO & India’s Space Missions: launch vehicles, Chandrayaan, Gaganyaan, Aditya-L1 and upcoming programmes.
- 8. Space Exploration: planetary science, human exploration, Moon and Mars missions, asteroid exploration, planetary defence and future missions.
- 9. Basic Physics for Space Science: mechanics, gravitation, electromagnetism, optics, waves, thermal and introductory modern physics.
- 10. Scientific Reasoning & Logical Aptitude: methodology, data interpretation, design, quantitative reasoning and advanced problem solving.
- 11. General Space Awareness: discoveries, international programmes, commercial space, sustainability and current affairs.

## Common Constants and Relations

| Quantity / relation                    | Value / expression                             |
|----------------------------------------|------------------------------------------------|
| Speed of light, $c$                    | $3.00 \times 10^8 \text{ m/s}$                 |
| Planck constant, $h$                   | $6.626 \times 10^{-34} \text{ J s}$            |
| Gravitational constant, $G$            | $6.67 \times 10^{-11} \text{ SI}$              |
| Earth gravitational parameter, $\mu_E$ | $3.986 \times 10^{14} \text{ m}^3/\text{s}^2$  |
| Solar mass, $M_{\text{sun}}$           | $1.99 \times 10^{30} \text{ kg}$               |
| Solar radius, $R_{\text{sun}}$         | $6.96 \times 10^8 \text{ m}$                   |
| 1 AU                                   | $1.496 \times 10^{11} \text{ m}$               |
| 1 parsec                               | $3.086 \times 10^{16} \text{ m}$               |
| 1 radian                               | 206265 arcsecond                               |
| Standard gravity, $g_0$                | $9.8 \text{ m/s}^2$                            |
| Wien constant                          | $2.898 \times 10^{-3} \text{ m K}$             |
| Photon relation                        | $E = hc/\lambda$ ; $hc = 1240 \text{ eV nm}$   |
| Circular orbit                         | $v = \sqrt{\mu/r}$ , $T = 2\pi \sqrt{r^3/\mu}$ |
| Escape speed                           | $v_{\text{esc}} = \sqrt{2\mu/r}$               |
| Rocket equation                        | $\Delta v = I_{\text{sp}} g_0 \ln(m_0/m_f)$    |
| Magnitude-flux relation                | $F_1/F_2 = 10^{(0.4 \Delta m)}$                |

# MOCK PAPER 1

## Advanced Level (Classes XI-XII) - 50 Questions - 200 Marks

|                 |       |             |  |
|-----------------|-------|-------------|--|
| Name            |       | Roll Number |  |
| School / Centre |       | Date        |  |
| Score           | / 200 | Time Taken  |  |

Instructions: Section A (Q1-Q40) contains single-correct MCQs: +4 for correct and -1 for incorrect. Section B (Q41-Q50) contains numerical-answer questions: +4 for correct and no negative marking. Use the common constants unless another value is specified.

### SECTION A - SINGLE-CORRECT MCQs

**1. Which instrument is primarily used to separate incoming astronomical light into its component wavelengths?**

- A. Spectrograph
- B. Photometer
- C. Coronagraph
- D. Star tracker

**2. Aditya-L1 is designed for nearly continuous solar observations from the vicinity of which point?**

- A. Sun-Earth L4
- B. Sun-Earth L1
- C. Earth-Moon L2
- D. A low polar Earth orbit

**3. Which one of the following is a terrestrial planet?**

- A. Saturn
- B. Jupiter
- C. Mars
- D. Neptune

**4. A geostationary satellite must have which combination of properties?**

- A. Circular orbit at any inclination, one sidereal-day period
- B. Polar orbit, 12 h period, westward motion
- C. Elliptical equatorial orbit, one solar-day period
- D. Equatorial circular orbit, one sidereal-day period, eastward motion

**5. According to Wien's displacement law, if the surface temperature of a star increases, the wavelength at which it emits most strongly will**

- A. decrease
- B. increase
- C. remain unchanged
- D. first increase and then decrease

**6. A star has an annual parallax of 0.125 arcsecond. Its distance is closest to**

- A. 12.5 pc
- B. 8 pc
- C. 125 pc
- D. 4 pc

**7. A star has apparent magnitude  $m = 10$  and absolute magnitude  $M = 5$ . Ignoring extinction, its distance is**

- A. 1000 pc
- B. 50 pc
- C. 100 pc
- D. 10 pc

**8. The Sun's diameter is approximately  $1.4 \times 10^9$  m and its distance from Earth is  $1.5 \times 10^{11}$  m. Its angular diameter is closest to**

- A. 53 degrees
- B. 5.3 degrees
- C. 0.053 degree
- D. 0.53 degree

**9. A telescope of aperture 0.20 m observes at wavelength 550 nm. Its Rayleigh diffraction limit is closest to**

- A. 0.69 arcsecond
- B. 0.07 arcsecond
- C. 6.9 arcseconds
- D. 69 arcseconds

**10. A 20 cm telescope requires 100 s to obtain a given signal-to-noise ratio for a bright source when detector and sky noise are negligible. Under the same conditions, a 50 cm telescope will require approximately**

- A. 6.4 s
- B. 16 s
- C. 40 s
- D. 250 s

**11. A hydrogen H-alpha line has laboratory wavelength 656.3 nm but is observed at 656.8 nm. The source's radial velocity is approximately**

- A. 23 km/s towards us
- B. 23 km/s away from us
- C. 230 km/s away from us
- D. 230 km/s towards us

**12. Star X has its blackbody peak at 500 nm and star Y at 1000 nm. The ratio  $T_X/T_Y$  is**

- A. 4
- B. 1
- C.  $1/2$
- D. 2

**13. Star A is 2.5 magnitudes brighter than star B. The flux received from A is approximately**

- A. 10 times that from B
- B. 100 times that from B
- C. 5 times that from B
- D. 2.5 times that from B

**14. A telescope of effective collecting area  $1.0 \text{ m}^2$  receives monochromatic flux  $3.0 \times 10^{-12} \text{ W m}^{-2}$  at 600 nm for 100 s. If the total detection efficiency is 50%, the number of detected photons is closest to**

- A.  $4.5 \times 10^5$
- B.  $4.5 \times 10^8$
- C.  $4.5 \times 10^6$
- D.  $4.5 \times 10^{10}$

**15. At an observatory, the local sidereal time is 10 h. A star has right ascension 8 h. Its hour angle is**

- A. 18 h east of the meridian
- B. 8 h west of the meridian

- C. 2 h west of the meridian
- D. 2 h east of the meridian

**16. An observer is at latitude 20 degrees N. A star of declination 10 degrees S crosses the local meridian. Its maximum altitude is**

- A. 50 degrees
- B. 30 degrees
- C. 80 degrees
- D. 60 degrees

**17. A satellite moves in a circular orbit of radius  $4R$  around a planet, where  $R$  is the planet's radius. Compared with the circular orbital speed just above the surface, its speed is**

- A. one-half
- B. four times
- C. twice
- D. one-fourth

**18. A small body orbits the Sun with semi-major axis 4 AU. Its orbital period is closest to**

- A. 16 years
- B. 8 years
- C. 2 years
- D. 4 years

**19. Treating Earth and Mars orbits as circular with radii 1.000 AU and 1.524 AU, the one-way time for an ideal Hohmann transfer is closest to**

- A. 365 days
- B. 129 days
- C. 259 days
- D. 189 days

**20. The escape speed from a planet's surface is  $v_e$ . At an altitude equal to the planet's radius above the surface, the escape speed is**

- A.  $v_e/2$
- B.  $2v_e$
- C.  $\sqrt{2} v_e$
- D.  $v_e/\sqrt{2}$

**21. A satellite of mass  $m$  is shifted from a circular orbit of radius  $r$  to a circular orbit of radius  $2r$  around a planet of mass  $M$ . The increase in its total mechanical energy is**

- A.  $GMm/(4r)$
- B.  $GMm/(8r)$
- C.  $3GMm/(4r)$
- D.  $GMm/(2r)$

**22. An exoplanet produces a transit depth of 1.0% across a uniformly bright star. The ratio of planet radius to stellar radius is**

- A. 0.05
- B. 0.10
- C. 0.01
- D. 0.20

**23. A planet orbits a one-solar-mass star once every 8 days. Assuming a circular orbit, its semi-major axis is closest to**

- A. 7.8 AU
- B. 0.78 AU

- C. 0.078 AU
- D. 0.008 AU

**24. For central transits across the same star and circular orbits, transit duration scales approximately as  $P^{1/3}$ . If orbital period becomes 8 times larger, the transit duration becomes**

- A. 16 times
- B. 4 times
- C. 8 times
- D. 2 times

**25. Two atmospheres have the same mean molecular mass. Planet B has twice the atmospheric temperature and four times the surface gravity of planet A. The ratio  $H_B/H_A$  of atmospheric scale heights is**

- A.  $1/2$
- B. 2
- C. 8
- D.  $1/8$

**26. A rocky exoplanet has mass 8 times Earth's mass and radius 2 times Earth's radius. Its mean density compared with Earth's is**

- A. four times
- B. the same
- C. one-half
- D. twice

**27. If the Earth-Moon separation were doubled while masses remained unchanged, the magnitude of the Moon's tide-generating effect on Earth would become approximately**

- A.  $1/2$
- B.  $1/16$
- C.  $1/8$
- D.  $1/4$

**28. A small Solar System body has perihelion 0.5 AU and aphelion 3.5 AU. Its orbital eccentricity is**

- A. 0.875
- B. 0.50
- C. 0.25
- D. 0.75

**29. A star has radius 2 times the Sun's radius and effective temperature half the Sun's. Its luminosity is**

- A.  $1/4$  solar
- B.  $1/2$  solar
- C.  $1/16$  solar
- D. 4 solar

**30. For main-sequence stars, suppose luminosity varies as  $M^{3.5}$  and available fuel varies as  $M$ . If the Sun's main-sequence lifetime is 10 billion years, a 2-solar-mass star has a lifetime closest to**

- A. 0.18 billion years
- B. 1.8 billion years
- C. 5.0 billion years
- D. 28 billion years

**31. Two stars have the same luminosity. Star B has twice the effective temperature of star A. The radius of B is**

- A. four times A
- B. one-half of A
- C. one-fourth of A
- D. one-sixteenth of A

**32. The pressure that mainly supports a typical white dwarf against gravitational collapse is**

- A. thermal gas pressure from hydrogen fusion
- B. neutron degeneracy pressure
- C. radiation pressure alone
- D. electron degeneracy pressure

**33. The Chandrasekhar limiting mass for a non-rotating white dwarf is approximately**

- A. 1.4 solar masses
- B. 0.14 solar mass
- C. 14 solar masses
- D. 140 solar masses

**34. Using the approximation that the Schwarzschild radius is 3 km per solar mass, the Schwarzschild radius of a 3-solar-mass object is**

- A. 1 km
- B. 9 km
- C. 3 km
- D. 27 km

**35. A binary star system has total mass 4 solar masses and orbital period 2 years. Its relative-orbit semi-major axis is closest to**

- A. 4.0 AU
- B. 1.6 AU
- C. 2.5 AU
- D. 1.0 AU

**36. For a Hubble constant of  $70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , the recession speed of a galaxy 200 Mpc away is**

- A. 3500 km/s
- B. 28000 km/s
- C. 7000 km/s
- D. 14000 km/s

**37. A galaxy has an approximately flat rotation curve at large radii. If circular speed  $v$  remains constant, the enclosed mass  $M(r)$  must vary approximately as**

- A.  $r$
- B.  $1/r$
- C. constant
- D.  $r^2$

**38. During cosmic expansion, the temperature of freely propagating cosmic microwave background radiation varies approximately as  $1/a$ , where  $a$  is the scale factor. If  $a$  doubles, the radiation temperature becomes**

- A. one-fourth
- B. one-half
- C. twice
- D. unchanged

**39. The magnetopause standoff distance scales approximately as  $P_{sw}^{-1/6}$ , where  $P_{sw}$  is solar-wind dynamic pressure. If  $P_{sw}$  increases by a factor of 64, the standoff distance becomes**

- A. one-eighth
- B. twice
- C. one-half
- D. one-fourth

**40. During a fine relative-navigation and docking experiment such as SpaDeX, reaction wheels can accumulate angular momentum and saturate. In low Earth orbit, which actuator can unload wheel momentum without consuming propellant?**

- A. Sun sensor
- B. High-gain antenna
- C. Star tracker
- D. Magnetorquer

## SECTION B - NUMERICAL ANSWER

**41. Using Earth's gravitational parameter  $3.986 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$  and a sidereal day of  $8.6164 \times 10^4 \text{ s}$ , calculate the geostationary orbital radius from Earth's centre, in kilometres. Give the answer to the nearest 100 km.**

Answer: \_\_\_\_\_

**42. For a diffraction-limited communication link at fixed wavelength, antenna gain is proportional to dish diameter squared. The transmitting dish diameter is doubled, the receiving dish diameter is halved, and the range is tripled. Find the ratio of new received power to old received power, correct to two decimal places.**

Answer: \_\_\_\_\_

**43. A single ideal rocket stage has specific impulse 300 s and must provide  $\Delta v = 5.4 \text{ km/s}$ . Take  $g_0 = 9.8 \text{ m/s}^2$ . Calculate the required initial-to-final mass ratio to two decimal places.**

Answer: \_\_\_\_\_

**44. A spacecraft in a circular orbit has speed 7.5 km/s. It begins a Hohmann transfer to a circular orbit of radius 4 times the initial radius. Calculate the first-burn  $\Delta v$  in km/s, to two decimal places.**

Answer: \_\_\_\_\_

**45. In a simplified model of a Gaganyaan air-drop test, a simulated crew module is released from 3.0 km altitude. Ignoring air resistance, calculate its speed just before reaching sea level. Take  $g = 9.8 \text{ m/s}^2$  and give the answer to the nearest m/s.**

Answer: \_\_\_\_\_

**46. Before the DART impact, Dimorphos completed one orbit in 11 h 55 min; afterwards, the period was 11 h 23 min. Calculate the percentage decrease in orbital period, to two decimal places.**

Answer: \_\_\_\_\_

**47. A star has radius 1.5 times solar and effective temperature 1.2 times solar. Calculate its luminosity in solar units, to two decimal places.**

Answer: \_\_\_\_\_

**48. A telescope must just resolve an angular separation of 0.50 arcsecond at wavelength 500 nm. Using Rayleigh's criterion, calculate the minimum aperture in metres, to three decimal places.**

Answer: \_\_\_\_\_

**49. An exoplanet causes a transit depth of 0.64% across a star of radius 1.25 solar radii. Take one Jupiter radius as 0.10 solar radius. Calculate the planet radius in Jupiter radii, to two decimal places.**

Answer: \_\_\_\_\_

**50. A detector records 620, 1120, 1620 and 2120 counts for exposure times 10, 20, 30 and 40 s. The detector has a fixed electronic bias of 100 counts and a dark-current rate of 12 counts/s. Assuming linear response, calculate the source count rate in counts/s.**

Answer: \_\_\_\_\_

# MOCK PAPER 2

## Advanced Level (Classes XI-XII) - 50 Questions - 200 Marks

|                 |       |             |  |
|-----------------|-------|-------------|--|
| Name            |       | Roll Number |  |
| School / Centre |       | Date        |  |
| Score           | / 200 | Time Taken  |  |

Instructions: Section A (Q1-Q40) contains single-correct MCQs: +4 for correct and -1 for incorrect. Section B (Q41-Q50) contains numerical-answer questions: +4 for correct and no negative marking. Use the common constants unless another value is specified.

### SECTION A - SINGLE-CORRECT MCQs

**1. Which instrument produces a spectrum that can reveal chemical composition and radial velocity of a star?**

- A. Radio altimeter
- B. Spectrograph
- C. Photometer
- D. Star tracker

**2. A solar observatory near the Sun-Earth L1 region is useful because it can**

- A. orbit the Moon every 24 hours
- B. remain fixed above one Indian longitude
- C. avoid all solar radiation
- D. view the Sun almost continuously before the solar wind reaches Earth

**3. A satellite image formed from reflected sunlight is an example of**

- A. passive remote sensing
- B. active radar ranging only
- C. orbital insertion
- D. attitude control

**4. Cryogenic rocket stages use propellants stored at very low temperature mainly because**

- A. low temperature makes gravity weaker
- B. cryogenic engines work without oxidiser
- C. the propellants are gases at ordinary temperature and must be liquefied for dense storage
- D. cold propellants remove the need for combustion

**5. A major observational reason for proposing dark matter in galaxies is that**

- A. black holes emit visible light from inside the event horizon
- B. outer stars orbit faster than expected from visible matter alone
- C. all galaxies emit identical spectra
- D. nearby galaxies show no redshift

**6. A star has annual parallax 0.05 arcsecond. Its distance is closest to**

- A. 10 pc
- B. 200 pc
- C. 40 pc
- D. 20 pc

**7. At an observatory the local sidereal time is 14 h and a star has right ascension 9 h. The star is**

- A. 5 h west of the meridian

- B. 5 h east of the meridian
- C. 14 h west of the meridian
- D. 9 h east of the meridian

**8. An observer at latitude 25 degrees observes a star of declination -15 degrees at upper meridian transit. Its maximum altitude is**

- A. 40 degrees
- B. 48 degrees
- C. 50 degrees
- D. 60 degrees

**9. A diffraction-limited telescope of aperture 0.3 m observes at wavelength 600 nm. Its Rayleigh angular resolution is closest to**

- A. 5.03 arcsecond
- B. 0.5 arcsecond
- C. 50.33 arcsecond
- D. 0.05 arcsecond

**10. A spectral line has laboratory wavelength 500 nm and is observed at 500.4 nm. For non-relativistic motion, the radial velocity is closest to**

- A. 24 km/s away from us
- B. 2400 km/s away from us
- C. 240 km/s towards us
- D. 240 km/s away from us

**11. Star A is 1.5 magnitudes brighter than star B. The flux ratio  $F_A/F_B$  is closest to**

- A. 3.98
- B. 39.81
- C. 1.99
- D. 1.5

**12. A telescope of collecting area  $0.8 \text{ m}^2$  receives monochromatic flux  $2.0 \times 10^{-12} \text{ W m}^{-2}$  at 500 nm for 120 s. Detection efficiency is 40%. Detected photons are closest to**

- A. 19317838.8
- B. 1931783.9
- C. 193178388.2
- D. 1931783881.7

**13. A spectrograph resolves two lines near 600 nm separated by 0.12 nm. Its minimum resolving power is**

- A. 2500
- B. 5000
- C. 50000
- D. 500

**14. For a bright source in the photon-limited regime, a 0.25 m telescope needs 160 s for a target signal-to-noise ratio. A 0.5 m telescope needs approximately**

- A. 80 s
- B. 10 s
- C. 102.4 s
- D. 40 s

**15. A small body orbits a one-solar-mass star with semi-major axis 2.25 AU. Its orbital period is closest to**

- A. 3.38 years
- B. 2.25 years
- C. 5.06 years

D. 1.5 years

**16. A satellite is moved from a circular orbit of radius  $r$  to a circular orbit of radius  $3r$ . The increase in total mechanical energy is what fraction of  $GMm/r$ ?**

A. 0.667

B. 0.167

C. 0.333

D. 0.403

**17. An orbit has periapsis 0.8 AU and apoapsis 5.2 AU. Its eccentricity is**

A. 1.364

B. 0.733

C. 0.367

D. 0.267

**18. Two planets have orbital periods 1 years and 1.88 years. Their synodic period is closest to**

A. 1.88 years

B. 0.88 years

C. 1.44 years

D. 2.14 years

**19. Atmosphere B has temperature 1.5 times, gravity 3 times and mean molecular mass 0.5 times those of atmosphere A. The ratio  $H_B/H_A$  is**

A. 1

B. 1.14

C. 0.5

D. 2

**20. A planet has mass 10 times Earth and radius 2 times Earth. Its mean density relative to Earth is**

A. 2.5

B. 5

C. 1.25

D. 0.8

**21. An exoplanet transit depth is 0.81%. Assuming a uniformly bright star,  $R_p/R_{\text{star}}$  is**

A. 0.045

B. 0.09

C. 0.18

D. 0.008

**22. Ignoring greenhouse effects and taking the same albedo convention as Earth, a planet around a star of luminosity  $4 L_{\text{sun}}$  at 2 AU has equilibrium temperature closest to**

A. 556 K

B. 139 K

C. 417 K

D. 278 K

**23. A star has radius  $1.8 R_{\text{sun}}$  and effective temperature  $1.1 T_{\text{sun}}$ . Its luminosity is**

A.  $4.744 L_{\text{sun}}$

B.  $3.92 L_{\text{sun}}$

C.  $2.178 L_{\text{sun}}$

D.  $1.186 L_{\text{sun}}$

**24. Assume main-sequence luminosity  $L$  proportional to  $M^{3.5}$  and available fuel proportional to  $M$ . A 3 Msun star has lifetime closest to**

- A. 0.06 billion years
- B. 3.33 billion years
- C. 0.64 billion years
- D. 1.92 billion years

**25. A visual binary has relative-orbit semi-major axis 4 AU and period 4 years. Its total mass is closest to**

- A. 1 Msun
- B. 4 Msun
- C. 8 Msun
- D. 2 Msun

**26. Using 3 km per solar mass, the Schwarzschild radius of a 8 Msun object is**

- A. 48 km
- B. 8 km
- C. 72 km
- D. 24 km

**27. A spectral feature emitted at 400 nm is observed at 460 nm. Its redshift  $z$  is**

- A. 0.15
- B. 0.075
- C. 0.87
- D. 1.15

**28. At large radius a galaxy has nearly constant circular speed. The enclosed gravitating mass  $M(r)$  must then vary approximately as**

- A.  $r^2$
- B. constant
- C.  $r$
- D.  $1/r$

**29. If the distance to a tide-raising body becomes 2.5 times larger while masses are unchanged, the tide-generating effect becomes**

- A. 0.16 of original
- B. 0.064 of original
- C. 15.625 of original
- D. 0.4 of original

**30. Magnetopause standoff distance scales as solar-wind pressure $^{-1/6}$ . If pressure rises by a factor 64, the distance becomes**

- A. 8 of original
- B. 0.125 of original
- C. 0.016 of original
- D. 0.5 of original

**31. A spacecraft is 1,200,000 km from Earth. Neglecting processing time, the round-trip light-time is closest to**

- A. 8 s
- B. 4 s
- C. 0.07 s
- D. 16 s

**32. An Earth-observation camera at altitude 600 km has pixel pitch 5 micrometre and focal length 1.2 m. Approximate ground sampling distance is**

- A. 0.25 m/pixel

- B. 5 m/pixel
- C. 2.5 m/pixel
- D. 25 m/pixel

**33. A solar array of area  $12 \text{ m}^2$  receives  $1360 \text{ W/m}^2$ . Efficiency is 30% and incidence angle from the normal is 30 degrees. Electrical power is closest to**

- A. 4896 W
- B. 4240.1 W
- C. 8480.1 W
- D. 2120 W

**34. A reaction wheel of moment of inertia  $0.06 \text{ kg m}^2$  changes speed by  $120 \text{ rad/s}$  in 8 s. Average torque magnitude on the spacecraft is**

- A. 57.6 N m
- B. 0.004 N m
- C. 250 N m
- D. 0.9 N m

**35. An ideal rocket stage has specific impulse 320 s and must supply  $\Delta v$  4.8 km/s. The initial-to-final mass ratio is closest to**

- A. 4.62
- B. 9.24
- C. 1.53
- D. 2.31

**36. A rocket ejects propellant at  $250 \text{ kg/s}$  with effective exhaust speed  $3100 \text{ m/s}$ . The pressure-thrust contribution is 20000 N. Total thrust is**

- A. 795250 N
- B. 775000 N
- C. 795000 N
- D. 937750 N

**37. The SpaDeX technology demonstration is directly relevant to future missions that require**

- A. sample analysis inside a Mars rover
- B. rendezvous, docking and transfer between separately launched spacecraft
- C. launching without a rocket
- D. continuous solar imaging from L1 only

**38. A key space-sustainability reason for passivating a retired upper stage is to**

- A. turn it into a crewed station
- B. increase its orbital speed indefinitely
- C. make it geostationary automatically
- D. remove stored energy that could cause an explosion and create debris

**39. NASA's Europa Clipper mission is designed primarily to assess whether Europa**

- A. has environments that could be suitable for life beneath or within its ice shell
- B. can be moved into Earth orbit
- C. has rings identical to Saturn's
- D. contains no water in any form

**40. Light of wavelength 450 nm strikes a detector material of work function 2.1 eV. Maximum photoelectron kinetic energy is**

- A. 0.33 eV
- B. 2.76 eV
- C. 0.66 eV

D. 2.1 eV

## SECTION B - NUMERICAL ANSWER

41. Using  $\mu_E = 3.986 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$  and sidereal day  $T = 86164 \text{ s}$ , calculate the geostationary orbital radius from Earth's centre in km. Give the nearest 100 km.

Answer: \_\_\_\_\_

42. A star has parallax 0.08 arcsecond. Calculate its distance in parsec.

Answer: \_\_\_\_\_

43. What telescope aperture in metres is required to just resolve 0.4 arcsecond at wavelength 500 nm? Use Rayleigh criterion and give three decimal places.

Answer: \_\_\_\_\_

44. An ideal rocket stage has  $I_{sp}=310 \text{ s}$  and must provide  $\Delta v=5.2 \text{ km/s}$ . Find  $m_0/m_f$  to two decimals.

Answer: \_\_\_\_\_

45. Assuming circular heliocentric orbits of radii 1 AU and 1.524 AU, calculate the one-way Hohmann transfer time in days. Give nearest day.

Answer: \_\_\_\_\_

46. A transit depth is 1.21% across a star of radius  $0.9 R_{\text{sun}}$ . Taking  $1 R_j = 0.1 R_{\text{sun}}$ , find planet radius in  $R_j$  to two decimals.

Answer: \_\_\_\_\_

47. A star has radius  $2 R_{\text{sun}}$  and effective temperature  $0.8 T_{\text{sun}}$ . Find luminosity in  $L_{\text{sun}}$  to two decimals.

Answer: \_\_\_\_\_

48. A galaxy recedes at  $12600 \text{ km/s}$ . Using  $H_0=70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , estimate its distance in Mpc to one decimal place.

Answer: \_\_\_\_\_

49. A spacecraft solar array of area  $8 \text{ m}^2$  receives  $1360 \text{ W/m}^2$  at normal incidence with efficiency 28%. Find electrical energy produced in 6 h, in kWh to two decimals.

Answer: \_\_\_\_\_

50. A detector records counts [410, 1010, 1610, 2210] at exposure times [5, 15, 25, 35] s. Fixed bias is 110 counts and dark current is 8 counts/s. Assuming linear response, find source count rate in counts/s.

Answer: \_\_\_\_\_

# MOCK PAPER 3

## Advanced Level (Classes XI-XII) - 50 Questions - 200 Marks

|                 |       |             |  |
|-----------------|-------|-------------|--|
| Name            |       | Roll Number |  |
| School / Centre |       | Date        |  |
| Score           | / 200 | Time Taken  |  |

Instructions: Section A (Q1-Q40) contains single-correct MCQs: +4 for correct and -1 for incorrect. Section B (Q41-Q50) contains numerical-answer questions: +4 for correct and no negative marking. Use the common constants unless another value is specified.

### SECTION A - SINGLE-CORRECT MCQs

**1. Photometry of an astronomical source primarily measures**

- A. the exact mass of the source
- B. the direction of its magnetic field only
- C. brightness through defined wavelength bands
- D. the age of every atom in it

**2. A reaction wheel changes spacecraft attitude by**

- A. exchanging angular momentum with the spacecraft body
- B. measuring star positions only
- C. producing lift from an atmosphere
- D. changing the spacecraft mass

**3. The habitable zone around a star is the range of distances where a rocky planet could, with a suitable atmosphere, maintain**

- A. a one-day orbital period
- B. a magnetic field stronger than the star
- C. a permanent ring system
- D. liquid water on its surface

**4. The advantage of placing Aditya-L1 near the Sun-Earth L1 region is best described as**

- A. blocking all charged particles from the Sun
- B. long-duration solar viewing with in-situ sampling of the upstream space environment
- C. remaining fixed above the lunar south pole
- D. eliminating the need for communication with Earth

**5. A major effect of reusable launch systems on the commercial space sector is the possibility of**

- A. allowing payloads to ignore mass limits
- B. making orbital mechanics unnecessary
- C. lowering marginal launch cost by recovering and reusing hardware
- D. eliminating all mission risk

**6. A star has annual parallax 0.2 arcsecond. Its distance is closest to**

- A. 5 pc
- B. 2.5 pc
- C. 50 pc
- D. 10 pc

- 7. An object of diameter  $3.50 \times 10^6$  m is at distance  $3.84 \times 10^8$  m. Using the small-angle approximation, its angular diameter is closest to**
- A. 0.052 degree  
B. 52.223 degree  
C. 5.222 degree  
D. 0.522 degree
- 8. A diffraction-limited telescope of aperture 0.15 m observes at wavelength 450 nm. Its Rayleigh angular resolution is closest to**
- A. 75.49 arcsecond  
B. 0.75 arcsecond  
C. 7.55 arcsecond  
D. 0.08 arcsecond
- 9. A spectral line has laboratory wavelength 656.3 nm and is observed at 655.9 nm. For non-relativistic motion, the radial velocity is closest to**
- A. 18.3 km/s towards us  
B. 182.8 km/s away from us  
C. 182.8 km/s towards us  
D. 1828.4 km/s towards us
- 10. Star A is 5 magnitudes brighter than star B. The flux ratio  $F_A/F_B$  is closest to**
- A. 100  
B. 5  
C. 1000  
D. 50
- 11. A telescope of collecting area  $1.5 \text{ m}^2$  receives monochromatic flux  $8.0 \times 10^{-13} \text{ W m}^{-2}$  at 800 nm for 200 s. Detection efficiency is 60%. Detected photons are closest to**
- A. 5795351.6  
B. 57953516.5  
C. 5795351645  
D. 579535164.5
- 12. A spectrograph resolves two lines near 500 nm separated by 0.05 nm. Its minimum resolving power is**
- A. 1000  
B. 10000  
C. 100000  
D. 5000
- 13. An observer at latitude  $-20^\circ$  observes a star of declination  $-50^\circ$  at upper meridian transit. Its maximum altitude is**
- A. 50 degrees  
B. 70 degrees  
C. 60 degrees  
D. 30 degrees
- 14. A small body orbits a one-solar-mass star with semi-major axis 6.25 AU. Its orbital period is closest to**
- A. 15.62 years  
B. 39.06 years  
C. 2.5 years  
D. 6.25 years
- 15. The escape speed at distance  $2R$  from a planet centre, compared with the surface escape speed at  $R$ , is**
- A. 2

- B. 1.414
- C. 0.5
- D. 0.707

**16. A satellite is moved from a circular orbit of radius  $r$  to a circular orbit of radius  $4r$ . The increase in total mechanical energy is what fraction of  $GMm/r$ ?**

- A. 0.188
- B. 0.375
- C. 0.25
- D. 0.625

**17. An orbit has periapsis 1.2 AU and apoapsis 4.8 AU. Its eccentricity is**

- A. 0.4
- B. 1.667
- C. 0.6
- D. 0.3

**18. Two planets have orbital periods 1 years and 0.615 years. Their synodic period is closest to**

- A. 1.6 years
- B. 0.61 years
- C. 0.81 years
- D. 0.39 years

**19. Atmosphere B has temperature 2 times, gravity 4 times and mean molecular mass 1 times those of atmosphere A. The ratio  $H_B/H_A$  is**

- A. 1
- B. 0.25
- C. 2
- D. 0.5

**20. An exoplanet transit depth is 2.25%. Assuming a uniformly bright star,  $R_p/R_{\text{star}}$  is**

- A. 0.022
- B. 0.15
- C. 0.3
- D. 0.075

**21. Ignoring greenhouse effects and taking the same albedo convention as Earth, a planet around a star of luminosity  $0.25 L_{\text{sun}}$  at 0.5 AU has equilibrium temperature closest to**

- A. 417 K
- B. 139 K
- C. 278 K
- D. 556 K

**22. A star has radius  $0.5 R_{\text{sun}}$  and effective temperature  $2 T_{\text{sun}}$ . Its luminosity is**

- A.  $4 L_{\text{sun}}$
- B.  $1.28 L_{\text{sun}}$
- C.  $2 L_{\text{sun}}$
- D.  $1 L_{\text{sun}}$

**23. Assume main-sequence luminosity  $L$  proportional to  $M^4$  and available fuel proportional to  $M$ . A  $5 M_{\text{sun}}$  star has lifetime closest to**

- A. 0.24 billion years
- B. 0.01 billion years
- C. 2 billion years
- D. 0.08 billion years

**24. A visual binary has relative-orbit semi-major axis 3 AU and period 1.5 years. Its total mass is closest to**

- A. 24  $M_{\text{sun}}$
- B. 12  $M_{\text{sun}}$
- C. 2  $M_{\text{sun}}$
- D. 6  $M_{\text{sun}}$

**25. Using 3 km per solar mass, the Schwarzschild radius of a 20  $M_{\text{sun}}$  object is**

- A. 20 km
- B. 180 km
- C. 60 km
- D. 120 km

**26. For  $H_0 = 68 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , a galaxy at 250 Mpc has Hubble recession speed**

- A. 17000 km/s
- B. 34000 km/s
- C. 318 km/s
- D. 8500 km/s

**27. A spectral feature emitted at 121.6 nm is observed at 486.4 nm. Its redshift  $z$  is**

- A. 1.5
- B. 4
- C. 0.25
- D. 3

**28. If the distance to a tide-raising body becomes 3 times larger while masses are unchanged, the tide-generating effect becomes**

- A. 0.333 of original
- B. 0.037 of original
- C. 27 of original
- D. 0.111 of original

**29. A moon has surface gravity 0.16 times Earth and radius 0.27 times Earth. Its escape speed relative to Earth is**

- A. 0.77
- B. 0.593
- C. 0.208
- D. 0.043

**30. Magnetopause standoff distance scales as solar-wind pressure <sup>$^{-1/6}$</sup> . If pressure rises by a factor 729, the distance becomes**

- A. 0.333 of original
- B. 0.037 of original
- C. 27 of original
- D. 0.001 of original

**31. At fixed wavelength, transmitting antenna diameter changes by factor 2, receiving diameter by 0.75, and range by 4. The received-power ratio new/old is**

- A. 7.111
- B. 0.375
- C. 0.07
- D. 0.141

**32. A spacecraft records data continuously at 12 Mbit/s for 3 h. Required storage, ignoring overhead, is**

- A. 2.02 GB
- B. 16.2 GB

C. 129.6 GB

D. 36 GB

**33. A reaction wheel of moment of inertia  $0.025 \text{ kg m}^2$  changes speed by  $400 \text{ rad/s}$  in  $20 \text{ s}$ . Average torque magnitude on the spacecraft is**

A.  $200 \text{ N m}$

B.  $800 \text{ N m}$

C.  $0.5 \text{ N m}$

D.  $0.001 \text{ N m}$

**34. A rocket produces thrust  $900000 \text{ N}$  with propellant flow  $300 \text{ kg/s}$ . Its effective specific impulse is closest to**

A.  $306.1 \text{ s}$

B.  $3000 \text{ s}$

C.  $612.2 \text{ s}$

D.  $31.2 \text{ s}$

**35. Two ideal stages have individual mass ratios 4 and 3.5. Neglecting interstage mass, the overall initial-to-final mass ratio is**

A. 3.74

B. 7.5

C. 0.5

D. 14

**36. The main purpose of a gravity turn after launch is to**

A. remove the need for orbital velocity

B. use gravity to bend the trajectory while reducing steering losses

C. keep the rocket vertically above the launch pad

D. increase atmospheric drag deliberately

**37. In a Gaganyaan integrated air-drop test, the key system validated by releasing a simulated crew module and recovering it at sea is the**

A. solar coronagraph

B. lunar drilling system

C. parachute-based deceleration and recovery sequence

D. cryogenic upper-stage restart

**38. For Europa Clipper, repeated close flybys rather than landing are intended to build global information about Europa's**

A. ice shell, composition, geology and possible subsurface ocean

B. surface cities and vegetation

C. lack of any magnetic interaction

D. ability to orbit Earth

**39. To test how detector temperature affects dark current, the best experimental design is to**

A. use one measurement at one temperature only

B. change temperature, exposure and illumination together

C. subtract all measurements before recording them

D. vary detector temperature while keeping exposure time and illumination fixed

**40. An astronomical telescope has objective focal length  $180 \text{ cm}$  and eyepiece focal length  $6 \text{ cm}$ . Its angular magnification magnitude in normal adjustment is**

A. 0

B. 30

C. 186

D. 174

**SECTION B - NUMERICAL ANSWER**

41. A spherical body has gravitational parameter  $4.905 \times 10^{12} \text{ m}^3/\text{s}^2$  and radius 1737 km. Calculate surface escape speed in km/s to two decimals.

Answer: \_\_\_\_\_

42. A satellite orbits a planet with  $\mu = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$  in a circular orbit of radius 7000 km from the centre. Calculate period in minutes to one decimal place.

Answer: \_\_\_\_\_

43. A line at laboratory wavelength 600 nm is observed at 600.6 nm. Calculate radial velocity in km/s to one decimal place; use positive for recession.

Answer: \_\_\_\_\_

44. Using Wien constant  $2.898 \times 10^{-3} \text{ m K}$ , find the peak wavelength in nm of a blackbody at 7250 K. Give nearest nm.

Answer: \_\_\_\_\_

45. Calculate atmospheric scale height in km for  $T = 250 \text{ K}$ , mean molecular mass 29 u and  $g = 9.8 \text{ m/s}^2$ . Give one decimal place.

Answer: \_\_\_\_\_

46. A spacecraft with moment of inertia  $1200 \text{ kg m}^2$  must change angular rate by  $0.004 \text{ rad/s}$  in 20 s. Find average torque in N m to three decimals.

Answer: \_\_\_\_\_

47. A kinetic impactor of mass 600 kg hits at 11 km/s. If ejecta enhance momentum by  $\beta = 2.4$ , calculate effective delivered momentum in units of  $10^9 \text{ kg m/s}$  to two decimals.

Answer: \_\_\_\_\_

48. A probe is 1.5 AU from Earth. Find one-way light time in minutes to one decimal place.

Answer: \_\_\_\_\_

49. Calculate photon energy in eV for wavelength 310 nm, using  $hc = 1240 \text{ eV nm}$ . Give two decimals.

Answer: \_\_\_\_\_

50. For data  $x = [1, 2, 3, 4, 5]$  and  $y = [2.1, 4.0, 6.2, 7.9, 10.1]$ , calculate the least-squares slope of y on x to two decimals.

Answer: \_\_\_\_\_

# MOCK PAPER 4

## Advanced Level (Classes XI-XII) - 50 Questions - 200 Marks

|                 |       |             |  |
|-----------------|-------|-------------|--|
| Name            |       | Roll Number |  |
| School / Centre |       | Date        |  |
| Score           | / 200 | Time Taken  |  |

Instructions: Section A (Q1-Q40) contains single-correct MCQs: +4 for correct and -1 for incorrect. Section B (Q41-Q50) contains numerical-answer questions: +4 for correct and no negative marking. Use the common constants unless another value is specified.

### SECTION A - SINGLE-CORRECT MCQs

**1. A microwave radar satellite that sends pulses and records their echoes performs**

- A. celestial photometry
- B. passive optical imaging only
- C. orbital insertion
- D. active remote sensing

**2. Gravitational lensing by a galaxy cluster can indicate dark matter because the inferred lensing mass may**

- A. depend only on the colour of one foreground star
- B. be independent of gravity
- C. greatly exceed the mass estimated from luminous matter
- D. always equal zero

**3. In a liquid-hydrogen/liquid-oxygen stage, the high specific impulse is mainly associated with**

- A. zero propellant mass
- B. high exhaust velocity produced by energetic combustion and low-molecular-mass products
- C. operation without a nozzle
- D. the absence of an oxidiser

**4. In the heliocentric frame, a planetary gravity assist can change a spacecraft's speed because**

- A. the spacecraft exchanges energy and momentum with the moving planet
- B. gravity creates energy from nothing
- C. the planet stops orbiting during the encounter
- D. the spacecraft burns propellant inside the planet

**5. The Kessler-syndrome concern in crowded orbital regions is that**

- A. radio signals remove debris automatically
- B. all debris immediately falls vertically to Earth
- C. satellites stop obeying gravity
- D. collisions can create fragments that trigger further collisions

**6. A star has apparent magnitude  $m = 15$  and absolute magnitude  $M = 5$ . Ignoring extinction, its distance is closest to**

- A. 10000 pc
- B. 100 pc
- C. 1000 pc
- D. 500 pc

**7. At an observatory the local sidereal time is 4 h and a star has right ascension 20 h. The star is**

- A. 4 h west of the meridian

- B. 8 h west of the meridian
- C. 8 h east of the meridian
- D. 20 h east of the meridian

**8. An observer at latitude 35 degrees observes a star of declination 65 degrees at upper meridian transit. Its maximum altitude is**

- A. 60 degrees
- B. 30 degrees
- C. 50 degrees
- D. 70 degrees

**9. An object of diameter  $1.39 \times 10^9$  m is at distance  $1.50 \times 10^{11}$  m. Using the small-angle approximation, its angular diameter is closest to**

- A. 0.053 degree
- B. 5.324 degree
- C. 53.236 degree
- D. 0.532 degree

**10. A diffraction-limited telescope of aperture 0.5 m observes at wavelength 550 nm. Its Rayleigh angular resolution is closest to**

- A. 0.03 arcsecond
- B. 27.68 arcsecond
- C. 0.28 arcsecond
- D. 2.77 arcsecond

**11. A spectral line has laboratory wavelength 434 nm and is observed at 435.3 nm. For non-relativistic motion, the radial velocity is closest to**

- A. 898.6 km/s towards us
- B. 898.6 km/s away from us
- C. 89.9 km/s away from us
- D. 8986.2 km/s away from us

**12. Star A is 0.75 magnitudes brighter than star B. The flux ratio  $F_A/F_B$  is closest to**

- A. 2
- B. 19.95
- C. 0.75
- D. 1

**13. A spectrograph resolves two lines near 656.3 nm separated by 0.02 nm. Its minimum resolving power is**

- A. 16408
- B. 3282
- C. 328150
- D. 32815

**14. A small body orbits a one-solar-mass star with semi-major axis 0.25 AU. Its orbital period is closest to**

- A. 0.5 years
- B. 0.25 years
- C. 0.12 years
- D. 0.06 years

**15. The escape speed at distance 5R from a planet centre, compared with the surface escape speed at R, is**

- A. 2.236
- B. 0.447
- C. 5
- D. 0.2

**16. A satellite is moved from a circular orbit of radius  $r$  to a circular orbit of radius  $2.5r$ . The increase in total mechanical energy is what fraction of  $GMm/r$ ?**

- A. 0.3
- B. 0.15
- C. 0.7
- D. 0.4

**17. An orbit has periapsis 0.3 AU and apoapsis 4.7 AU. Its eccentricity is**

- A. 1.136
- B. 0.44
- C. 0.12
- D. 0.88

**18. Two planets have orbital periods 1 years and 11.86 years. Their synodic period is closest to**

- A. 10.86 years
- B. 11.86 years
- C. 1.09 years
- D. 6.43 years

**19. Atmosphere B has temperature 0.8 times, gravity 0.4 times and mean molecular mass 2 times those of atmosphere A. The ratio  $H_B/H_A$  is**

- A. 0.5
- B. 1
- C. 1.14
- D. 2

**20. An exoplanet transit depth is 0.36%. Assuming a uniformly bright star,  $R_p/R_{\text{star}}$  is**

- A. 0.06
- B. 0.12
- C. 0.03
- D. 0.004

**21. Ignoring greenhouse effects and taking the same albedo convention as Earth, a planet around a star of luminosity  $2.25 L_{\text{sun}}$  at 1.5 AU has equilibrium temperature closest to**

- A. 139 K
- B. 556 K
- C. 417 K
- D. 278 K

**22. A star has radius  $3 R_{\text{sun}}$  and effective temperature  $0.7 T_{\text{sun}}$ . Its luminosity is**

- A.  $1.47 L_{\text{sun}}$
- B.  $0.54 L_{\text{sun}}$
- C.  $2.161 L_{\text{sun}}$
- D.  $4.41 L_{\text{sun}}$

**23. Assume main-sequence luminosity  $L$  proportional to  $M^3$  and available fuel proportional to  $M$ . A  $2.5 M_{\text{sun}}$  star has lifetime closest to**

- A. 4.8 billion years
- B. 1.6 billion years
- C. 0.16 billion years
- D. 4 billion years

**24. A visual binary has relative-orbit semi-major axis 5 AU and period 5 years. Its total mass is closest to**

- A.  $5 M_{\text{sun}}$
- B.  $1 M_{\text{sun}}$

C. 2.5 Msun

D. 10 Msun

**25. Using 3 km per solar mass, the Schwarzschild radius of a  $1e+06$  Msun object is**

A. 6000000 km

B. 1000000 km

C. 9000000 km

D. 3000000 km

**26. For  $H_0 = 72 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , a galaxy at 80 Mpc has Hubble recession speed**

A. 11520 km/s

B. 152 km/s

C. 5760 km/s

D. 2880 km/s

**27. A spectral feature emitted at 500 nm is observed at 750 nm. Its redshift  $z$  is**

A. 1.5

B. 0.5

C. 0.667

D. 0.25

**28. The observation that distant Type Ia supernovae are dimmer than expected in a steadily decelerating universe is interpreted as evidence for**

A. accelerated cosmic expansion associated conceptually with dark energy

B. the absence of all galaxies beyond the Milky Way

C. a constant scale factor

D. no redshift in distant galaxies

**29. If the distance to a tide-raising body becomes 1.5 times larger while masses are unchanged, the tide-generating effect becomes**

A. 3.375 of original

B. 0.444 of original

C. 0.667 of original

D. 0.296 of original

**30. Magnetopause standoff distance scales as solar-wind pressure $^{-1/6}$ . If pressure rises by a factor 4, the distance becomes**

A. 0.25 of original

B. 0.5 of original

C. 0.794 of original

D. 2 of original

**31. A spacecraft is 384,400 km from Earth. Neglecting processing time, the round-trip light-time is closest to**

A. 1.28 s

B. 2.56 s

C. 0.02 s

D. 5.13 s

**32. An Earth-observation camera at altitude 500 km has pixel pitch 10 micrometre and focal length 0.8 m. Approximate ground sampling distance is**

A. 6.25 m/pixel

B. 62.5 m/pixel

C. 12.5 m/pixel

D. 0.62 m/pixel

- 33. A spacecraft records data continuously at 6 Mbit/s for 12 h. Required storage, ignoring overhead, is**  
 A. 259.2 GB  
 B. 4.05 GB  
 C. 72 GB  
 D. 32.4 GB
- 34. At fixed wavelength, transmitting antenna diameter changes by factor 1.5, receiving diameter by 1.5, and range by 2. The received-power ratio new/old is**  
 A. 0.79  
 B. 1.125  
 C. 1.266  
 D. 0.633
- 35. An ideal rocket stage has specific impulse 450 s and must supply  $\Delta v$  7.5 km/s. The initial-to-final mass ratio is closest to**  
 A. 10.96  
 B. 5.48  
 C. 1.7  
 D. 2.74
- 36. A rocket ejects propellant at 180 kg/s with effective exhaust speed 4200 m/s. The pressure-thrust contribution is -5000 N. Total thrust is**  
 A. 751000 N  
 B. 751180 N  
 C. 756000 N  
 D. 914760 N
- 37. A kinetic impactor of mass 500 kg strikes an asteroid at 12 km/s. Ignoring ejecta enhancement, delivered momentum is**  
 A. 6000 kg m/s  
 B. 6000000000 kg m/s  
 C. 600000 kg m/s  
 D. 6000000 kg m/s
- 38. Light of wavelength 400 nm strikes a detector material of work function 2 eV. Maximum photoelectron kinetic energy is**  
 A. 3.1 eV  
 B. 2 eV  
 C. 1.1 eV  
 D. 0.55 eV
- 39. If  $z=xy$ , with independent percentage uncertainties 3% in  $x$  and 4% in  $y$ , the maximum percentage uncertainty in  $z$  is**  
 A. 0.1%  
 B. 7%  
 C. 5%  
 D. 1%
- 40. Which pairing correctly represents the two broad measurement modes used by Aditya-L1?**  
 A. Remote imaging/spectroscopy of the Sun and in-situ particle/field measurements  
 B. Radar mapping of Venus and ocean altimetry of Earth  
 C. Lunar drilling and Mars sample return  
 D. Crewed docking and biological life support

**SECTION B - NUMERICAL ANSWER**

41. Using  $\mu_E = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$ , sidereal day 86164 s and Earth radius 6378 km, calculate geostationary altitude above the equator in km. Give nearest 100 km.

Answer: \_\_\_\_\_

42. Assuming circular heliocentric orbits of radii 1 AU and 2.77 AU, calculate the one-way Hohmann transfer time in days. Give nearest day.

Answer: \_\_\_\_\_

43. A transit depth is 0.49% across a star of radius 1.4  $R_{\text{sun}}$ . Taking  $1 R_j = 0.1 R_{\text{sun}}$ , find planet radius in  $R_j$  to two decimals.

Answer: \_\_\_\_\_

44. A star has radius 0.8  $R_{\text{sun}}$  and effective temperature 1.5  $T_{\text{sun}}$ . Find luminosity in  $L_{\text{sun}}$  to two decimals.

Answer: \_\_\_\_\_

45. A galaxy recedes at 18000 km/s. Using  $H_0 = 72 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , estimate its distance in Mpc to one decimal place.

Answer: \_\_\_\_\_

46. A spacecraft solar array of area  $15 \text{ m}^2$  receives  $1000 \text{ W/m}^2$  at normal incidence with efficiency 25%. Find electrical energy produced in 8 h, in kWh to two decimals.

Answer: \_\_\_\_\_

47. An ideal rocket stage has  $I_{\text{sp}} = 450 \text{ s}$  and must provide  $\Delta v = 8.2 \text{ km/s}$ . Find  $m_0/m_f$  to two decimals.

Answer: \_\_\_\_\_

48. A spherical body has gravitational parameter  $1.267 \times 10^{17} \text{ m}^3/\text{s}^2$  and radius 71492 km. Calculate surface escape speed in km/s to two decimals.

Answer: \_\_\_\_\_

49. Calculate atmospheric scale height in km for  $T = 165 \text{ K}$ , mean molecular mass 2.3 u and  $g = 3.7 \text{ m/s}^2$ . Give one decimal place.

Answer: \_\_\_\_\_

50. A detector records counts [700, 1250, 1800, 2350] at exposure times [10, 20, 30, 40] s. Fixed bias is 150 counts and dark current is 5 counts/s. Assuming linear response, find source count rate in counts/s.

Answer: \_\_\_\_\_

# MOCK PAPER 5

## Advanced Level (Classes XI-XII) - 50 Questions - 200 Marks

|                 |       |             |  |
|-----------------|-------|-------------|--|
| Name            |       | Roll Number |  |
| School / Centre |       | Date        |  |
| Score           | / 200 | Time Taken  |  |

Instructions: Section A (Q1-Q40) contains single-correct MCQs: +4 for correct and -1 for incorrect. Section B (Q41-Q50) contains numerical-answer questions: +4 for correct and no negative marking. Use the common constants unless another value is specified.

### SECTION A - SINGLE-CORRECT MCQs

**1. When the aim is to measure the wavelength separation and relative strength of stellar absorption lines, the most suitable attachment to a telescope is a**

- A. spectrograph
- B. reaction wheel
- C. bolometer without filters
- D. radio transponder

**2. A planet lying inside a star's nominal habitable zone is not automatically habitable because surface conditions also depend strongly on**

- A. the alphabetical name of the star
- B. whether it has exactly one moon
- C. its atmosphere and planetary environment
- D. whether its year equals one Earth year

**3. If a spacecraft accelerates an internal reaction wheel clockwise, the spacecraft body initially tends to rotate**

- A. clockwise with twice the angular momentum
- B. anticlockwise
- C. only if atmospheric lift is present
- D. without any angular response

**4. The autonomous second docking and inter-satellite power-transfer demonstration of SpaDeX most directly tested**

- A. solar spectroscopy at L1
- B. radio astronomy from the lunar far side
- C. Mars atmospheric entry
- D. close-proximity guidance, docking interfaces and post-docking electrical transfer

**5. For a commercial reusable launch system, economic benefit depends not only on recovery but also on**

- A. refurbishment cost, reliability and achievable flight rate
- B. making payload mass irrelevant
- C. violating the rocket equation
- D. removing all ground operations

**6. A star has annual parallax 0.04 arcsecond. Its distance is closest to**

- A. 50 pc
- B. 250 pc
- C. 25 pc
- D. 12.5 pc

**7. At an observatory the local sidereal time is 7.5 h and a star has right ascension 10.25 h. The star is**

- A. 10.25 h west of the meridian
- B. 2.75 h east of the meridian
- C. 7.5 h east of the meridian
- D. 2.75 h west of the meridian

**8. An observer at latitude  $-30^\circ$  observes a star of declination  $20^\circ$  at upper meridian transit. Its maximum altitude is**

- A.  $50^\circ$
- B.  $64^\circ$
- C.  $30^\circ$
- D.  $40^\circ$

**9. A diffraction-limited telescope of aperture 2 m observes at wavelength 1000 nm. Its Rayleigh angular resolution is closest to**

- A. 0.13 arcsecond
- B. 1.26 arcsecond
- C. 0.01 arcsecond
- D. 12.58 arcsecond

**10. A spectral line has laboratory wavelength 486.1 nm and is observed at 487.6 nm. For non-relativistic motion, the radial velocity is closest to**

- A. 92.6 km/s away from us
- B. 9257.4 km/s away from us
- C. 925.7 km/s away from us
- D. 925.7 km/s towards us

**11. Star A is 3.2 magnitudes brighter than star B. The flux ratio  $F_A/F_B$  is closest to**

- A. 3.2
- B. 19.05
- C. 9.53
- D. 190.55

**12. A telescope of collecting area  $2 \text{ m}^2$  receives monochromatic flux  $5.0 \times 10^{-14} \text{ W m}^{-2}$  at 400 nm for 600 s. Detection efficiency is 70%. Detected photons are closest to**

- A. 845155448.2
- B. 8451554.5
- C. 845155.4
- D. 84515544.8

**13. A spectrograph resolves two lines near 121.6 nm separated by 0.01 nm. Its minimum resolving power is**

- A. 12160
- B. 6080
- C. 1216
- D. 121600

**14. A small body orbits a one-solar-mass star with semi-major axis 10 AU. Its orbital period is closest to**

- A. 10 years
- B. 100 years
- C. 31.62 years
- D. 3.16 years

**15. The escape speed at distance  $1.44R$  from a planet centre, compared with the surface escape speed at  $R$ , is**

- A. 0.694
- B. 0.833

- C. 1.2
- D. 1.44

**16. A satellite is moved from a circular orbit of radius  $r$  to a circular orbit of radius  $10r$ . The increase in total mechanical energy is what fraction of  $GMm/r$ ?**

- A. 0.55
- B. 0.1
- C. 0.225
- D. 0.45

**17. An orbit has periapsis 0.2 AU and apoapsis 9.8 AU. Its eccentricity is**

- A. 0.96
- B. 0.48
- C. 1.042
- D. 0.04

**18. Two planets have orbital periods 0.241 years and 0.615 years. Their synodic period is closest to**

- A. 0.43 years
- B. 0.15 years
- C. 0.4 years
- D. 0.37 years

**19. Atmosphere B has temperature 3 times, gravity 6 times and mean molecular mass 0.25 times those of atmosphere A. The ratio  $H_B/H_A$  is**

- A. 0.5
- B. 2
- C. 1
- D. 4

**20. A planet has mass 20 times Earth and radius 2.2 times Earth. Its mean density relative to Earth is**

- A. 0.532
- B. 3.757
- C. 9.091
- D. 1.878

**21. An exoplanet transit depth is 0.04%. Assuming a uniformly bright star,  $R_p/R_{\text{star}}$  is**

- A. 0.02
- B. 0.04
- C. 0
- D. 0.01

**22. Ignoring greenhouse effects and taking the same albedo convention as Earth, a planet around a star of luminosity  $16 L_{\text{sun}}$  at 8 AU has equilibrium temperature closest to**

- A. 393 K
- B. 295 K
- C. 197 K
- D. 98 K

**23. A star has radius  $2.5 R_{\text{sun}}$  and effective temperature  $1.4 T_{\text{sun}}$ . Its luminosity is**

- A.  $4.9 L_{\text{sun}}$
- B.  $24.01 L_{\text{sun}}$
- C.  $12.25 L_{\text{sun}}$
- D.  $6.002 L_{\text{sun}}$

**24. Assume main-sequence luminosity  $L$  proportional to  $M^{3.5}$  and available fuel proportional to  $M$ . A 8 Msun star has lifetime closest to**

- A. 0.01 billion years
- B. 1.25 billion years
- C. 0.17 billion years
- D. 0.06 billion years

**25. A visual binary has relative-orbit semi-major axis 6 AU and period 3 years. Its total mass is closest to**

- A. 24 Msun
- B. 12 Msun
- C. 2 Msun
- D. 48 Msun

**26. Using 3 km per solar mass, the Schwarzschild radius of a  $4 \times 10^6$  Msun object is**

- A. 24000000 km
- B. 36000000 km
- C. 12000000 km
- D. 4000000 km

**27. For  $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , a galaxy at 500 Mpc has Hubble recession speed**

- A. 67400 km/s
- B. 33700 km/s
- C. 567 km/s
- D. 16850 km/s

**28. A spectral feature emitted at 400 nm is observed at 1000 nm. Its redshift  $z$  is**

- A. 0.75
- B. 2.5
- C. 0.4
- D. 1.5

**29. If the distance to a tide-raising body becomes 4 times larger while masses are unchanged, the tide-generating effect becomes**

- A. 0.016 of original
- B. 0.062 of original
- C. 64 of original
- D. 0.25 of original

**30. A moon has surface gravity 0.38 times Earth and radius 0.53 times Earth. Its escape speed relative to Earth is**

- A. 0.717
- B. 0.847
- C. 0.449
- D. 0.201

**31. Magnetopause standoff distance scales as solar-wind pressure $^{-1/6}$ . If pressure rises by a factor 15625, the distance becomes**

- A. 0.008 of original
- B. 0.2 of original
- C. 0 of original
- D. 125 of original

**32. An Earth-observation camera at altitude 700 km has pixel pitch 4 micrometre and focal length 2 m. Approximate ground sampling distance is**

- A. 0.14 m/pixel
- B. 2.8 m/pixel

- C. 14 m/pixel
- D. 1.4 m/pixel

**33. A solar array of area  $20 \text{ m}^2$  receives  $1360 \text{ W/m}^2$ . Efficiency is 32% and incidence angle from the normal is 60 degrees. Electrical power is closest to**

- A. 4352 W
- B. 2176 W
- C. 8704 W
- D. 11141.1 W

**34. A spacecraft records data continuously at 50 Mbit/s for 1.5 h. Required storage, ignoring overhead, is**

- A. 4.22 GB
- B. 270 GB
- C. 33.75 GB
- D. 75 GB

**35. A reaction wheel of moment of inertia  $0.08 \text{ kg m}^2$  changes speed by 250 rad/s in 5 s. Average torque magnitude on the spacecraft is**

- A. 0.002 N m
- B. 4 N m
- C. 625 N m
- D. 100 N m

**36. A rocket produces thrust  $1.5 \times 10^6 \text{ N}$  with propellant flow 500 kg/s. Its effective specific impulse is closest to**

- A. 31.2 s
- B. 612.2 s
- C. 3000 s
- D. 306.1 s

**37. Two ideal stages have individual mass ratios 5 and 4. Neglecting interstage mass, the overall initial-to-final mass ratio is**

- A. 20
- B. 4.47
- C. 1
- D. 9

**38. Aditya-L1 combines remote-sensing instruments with in-situ instruments mainly to study**

- A. the interior of Jupiter using seismology
- B. Earth communication traffic only
- C. the Sun and the solar wind near the Sun-Earth L1 region
- D. the geology of Mars from its surface

**39. Europa is a major target in planetary exploration because evidence suggests that beneath its icy surface it may contain**

- A. no water-bearing material
- B. a global salty ocean
- C. a second Sun
- D. a dense ring system larger than Saturn's

**40. An astronomical telescope has objective focal length 200 cm and eyepiece focal length 5 cm. Its angular magnification magnitude in normal adjustment is**

- A. 205
- B. 0
- C. 195
- D. 40

**SECTION B - NUMERICAL ANSWER**

41. A star has parallax 0.025 arcsecond. Calculate its distance in parsec.

Answer: \_\_\_\_\_

42. What telescope aperture in metres is required to just resolve 0.2 arcsecond at wavelength 650 nm? Use Rayleigh criterion and give three decimal places.

Answer: \_\_\_\_\_

43. Using Wien constant  $2.898 \times 10^{-3} \text{ m K}$ , find the peak wavelength in nm of a blackbody at 5800 K. Give nearest nm.

Answer: \_\_\_\_\_

44. A satellite orbits a planet with  $\mu = 4.283 \times 10^{13} \text{ m}^3/\text{s}^2$  in a circular orbit of radius 6000 km from the centre. Calculate period in minutes to one decimal place.

Answer: \_\_\_\_\_

45. An ideal rocket stage has  $I_{sp} = 380 \text{ s}$  and must provide  $\Delta v = 6.7 \text{ km/s}$ . Find  $m_0/m_f$  to two decimals.

Answer: \_\_\_\_\_

46. A spacecraft with moment of inertia  $2500 \text{ kg m}^2$  must change angular rate by  $0.006 \text{ rad/s}$  in 15 s. Find average torque in N m to three decimals.

Answer: \_\_\_\_\_

47. A kinetic impactor of mass 750 kg hits at 14 km/s. If ejecta enhance momentum by  $\beta = 3$ , calculate effective delivered momentum in units of  $10^9 \text{ kg m/s}$  to two decimals.

Answer: \_\_\_\_\_

48. A probe is 2.2 AU from Earth. Find one-way light time in minutes to one decimal place.

Answer: \_\_\_\_\_

49. Calculate photon energy in eV for wavelength 620 nm, using  $hc = 1240 \text{ eV nm}$ . Give two decimals.

Answer: \_\_\_\_\_

50. For data  $x = [0, 1, 2, 3, 4]$  and  $y = [1.0, 3.2, 5.1, 7.3, 9.2]$ , calculate the least-squares slope of y on x to two decimals.

Answer: \_\_\_\_\_

# Mock Paper 1 - Answer Key and Solutions

## Section A Answer Key

| 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. A  | 2. B  | 3. C  | 4. D  | 5. A  | 6. B  | 7. C  | 8. D  | 9. A  | 10. B |
| 11. C | 12. D | 13. A | 14. B | 15. C | 16. D | 17. A | 18. B | 19. C | 20. D |
| 21. A | 22. B | 23. C | 24. D | 25. A | 26. B | 27. C | 28. D | 29. A | 30. B |
| 31. C | 32. D | 33. A | 34. B | 35. C | 36. D | 37. A | 38. B | 39. C | 40. D |

## Section B Numerical Answers

| Q41-Q42               | Q43-Q44              | Q45-Q46             | Q47-Q48               | Q49-Q50            |
|-----------------------|----------------------|---------------------|-----------------------|--------------------|
| 41: 42200<br>42: 0.11 | 43: 6.28<br>44: 1.99 | 45: 242<br>46: 4.48 | 47: 4.67<br>48: 0.252 | 49: 1.00<br>50: 38 |

## Concise Solutions - Section A

- Answer: A.** A spectrograph disperses light by wavelength so that spectral lines and continua can be studied.
- Answer: B.** The Sun-Earth L1 region provides an uninterrupted view of the Sun while remaining gravitationally linked to the Earth-Sun system.
- Answer: C.** Mars is a rocky terrestrial planet; Jupiter, Saturn and Neptune are giant planets.
- Answer: D.** Only an eastward circular orbit in Earth's equatorial plane with a sidereal-day period remains fixed over one longitude.
- Answer: A.** Wien's law gives  $\lambda_{\max} T = \text{constant}$ , so the peak wavelength decreases as temperature rises.
- Answer: B.** Distance in parsecs is  $d = 1/p$ , where  $p$  is in arcseconds:  $d = 1/0.125 = 8$  pc.
- Answer: C.** Using  $m-M = 5 \log_{10}(d/10 \text{ pc})$ ,  $5 = 5 \log_{10}(d/10)$ , so  $d = 100$  pc.
- Answer: D.** For a small angle,  $\theta = D/d = 9.33 \times 10^{-3} \text{ rad} = 0.535$  degree.
- Answer: A.**  $\theta = 1.22 \lambda/D = 3.36 \times 10^{-6} \text{ rad} = 0.69$  arcsecond.
- Answer: B.** Collected photons scale as  $D^2 t$ . Therefore  $t_2/t_1 = (D_1/D_2)^2 = (20/50)^2 = 0.16$ .
- Answer: C.** For non-relativistic Doppler shift,  $v/c = \Delta \lambda/\lambda = 0.5/656.3$ , giving about 228 km/s. The longer wavelength is a redshift, so the source is receding.
- Answer: D.** Wien's law makes temperature inversely proportional to peak wavelength, so  $T_X/T_Y = 1000/500 = 2$ .
- Answer: A.** A magnitude difference of 2.5 corresponds to a flux factor  $10^{(0.4 \times 2.5)} = 10$ .
- Answer: B.** Detected energy is  $3.0 \times 10^{-12} \times 100 \times 0.5 = 1.5 \times 10^{-10} \text{ J}$ . A 600 nm photon has energy  $hc/\lambda = 3.31 \times 10^{-19} \text{ J}$ , giving about  $4.53 \times 10^8$  photons.
- Answer: C.** Hour angle  $H = \text{LST} - \text{RA} = 2$  h. Positive hour angle means the star has crossed the meridian and lies west of it.
- Answer: D.** At culmination, altitude =  $90^\circ - |\text{latitude} - \text{declination}| = 90 - |20 - (-10)| = 60$  degrees.
- Answer: A.** Circular speed  $v = \sqrt{GM/r}$ . Increasing  $r$  by a factor of 4 reduces  $v$  by a factor of 2.
- Answer: B.** For solar orbits in years and AU,  $P^2 = a^3$ . Thus  $P = \sqrt{64} = 8$  years.
- Answer: C.** The transfer ellipse has  $a = (1+1.524)/2 = 1.262$  AU. Its period is  $\sqrt{a^3} = 1.418$  years; half of this is about 259 days.
- Answer: D.** The distance from the centre is then  $2R$  and escape speed varies as  $r^{-1/2}$ , giving  $v/\sqrt{2}$ .
- Answer: A.** Circular-orbit energy is  $-GMm/(2r)$ . The change is  $-GMm/(4r) - [-GMm/(2r)] = GMm/(4r)$ .
- Answer: B.** Transit depth is approximately  $(R_p/R_s)^2$ , so  $R_p/R_s = \sqrt{0.01} = 0.10$ .
- Answer: C.** Using  $P^2 = a^3$  with  $P$  in years,  $P = 8/365.25$  year gives  $a = P^{2/3}$  about 0.078 AU.
- Answer: D.** Since duration is proportional to  $P^{1/3}$ , increasing  $P$  by 8 increases duration by  $8^{1/3} = 2$ .
- Answer: A.** Scale height  $H = kT/(mg)$ , so  $H_B/H_A = 2/4 = 1/2$ .
- Answer: B.** Density scales as  $M/R^3$ . The ratio is  $8/2^3 = 1$ .
- Answer: C.** The differential tidal force varies approximately as  $1/r^3$ , hence doubling  $r$  reduces it to  $1/8$ .
- Answer: D.**  $e = (r_a - r_p)/(r_a + r_p) = (3.5 - 0.5)/(3.5 + 0.5) = 3/4 = 0.75$ .

- 29. Answer: A.**  $L/L_{\text{sun}} = (R/R_{\text{sun}})^2 (T/T_{\text{sun}})^4 = 4 \times (1/2)^4 = 1/4$ .
- 30. Answer: B.** Lifetime scales as  $M/L = M^{-2.5}$ . Thus  $10/2^{2.5} = 1.77$  billion years.
- 31. Answer: C.** Since  $L$  is proportional to  $R^2 T^4$ , equal  $L$  gives  $R$  proportional to  $T^{-2}$ . Doubling  $T$  reduces radius by 4.
- 32. Answer: D.** White dwarfs are supported mainly by electron degeneracy pressure after ordinary fusion has ceased.
- 33. Answer: A.** The standard approximate Chandrasekhar limit is about 1.4 solar masses.
- 34. Answer: B.**  $R_s$  is proportional to mass; 3 solar masses gives about 9 km.
- 35. Answer: C.** In solar units,  $a^3 = MP^2 = 4 \times 4 = 16$ , so  $a = 16^{1/3} = 2.52$  AU.
- 36. Answer: D.** Hubble's law  $v = H_0 d$  gives  $70 \times 200 = 14000$  km/s.
- 37. Answer: A.** For circular motion,  $v^2 = GM(r)/r$ . Constant  $v$  therefore requires  $M(r)$  proportional to  $r$ .
- 38. Answer: B.** Cosmological redshift stretches wavelengths in proportion to  $a$ , so the blackbody temperature falls as  $1/a$ .
- 39. Answer: C.**  $64 = 2^6$ , so  $64^{(-1/6)} = 1/2$ .
- 40. Answer: D.** A magnetorquer interacts with Earth's magnetic field to generate an external torque and desaturate reaction wheels.

## Worked Solutions - Section B

- 41. Answer: 42200.** For a circular orbit,  $r = [\mu T^2 / (4 \pi^2)]^{1/3} = 4.2164 \times 10^7 \text{ m} = 42164 \text{ km}$ , which rounds to 42200 km.
- 42. Answer: 0.11.** Friis scaling gives  $P_r$  proportional to  $Dt^2 Dr^2/R^2$ . The diameter factors cancel:  $2^2 \times (1/2)^2 = 1$ . Tripling range gives  $1/9 = 0.111\dots$ , hence 0.11.
- 43. Answer: 6.28.** Tsiolkovsky's equation gives  $m_0/m_f = \exp[\Delta v / (I_{sp} g_0)] = \exp[5400 / (300 \times 9.8)] = 6.276$ , hence 6.28.
- 44. Answer: 1.99.** For  $r_2 = 4r_1$ , the transfer perigee speed is  $v_1 \sqrt{2r_2/(r_1+r_2)} = v_1 \sqrt{8/5}$ . Thus  $\Delta v_1 = 7.5[\sqrt{8/5}-1] = 1.99 \text{ km/s}$ .
- 45. Answer: 242.** Using  $v = \sqrt{2gh}$ ,  $v = \sqrt{2 \times 9.8 \times 3000} = 242.5 \text{ m/s}$ , which rounds to 242 m/s. The actual test uses a sequenced parachute system to avoid such a speed.
- 46. Answer: 4.48.** The periods are 715 min and 683 min. Percentage decrease =  $32/715 \times 100 = 4.4755\%$ , hence 4.48%.
- 47. Answer: 4.67.**  $L/L_{\text{sun}} = 1.5^2 \times 1.2^4 = 2.25 \times 2.0736 = 4.6656$ , hence 4.67.
- 48. Answer: 0.252.** Convert 0.50 arcsecond to radians:  $\theta = 0.50/206265$ . Then  $D = 1.22 \lambda / \theta = 0.2516 \text{ m}$ , hence 0.252 m.
- 49. Answer: 1.00.**  $R_p/R_s = \sqrt{0.0064} = 0.08$ . Thus  $R_p = 0.08 \times 1.25 = 0.10$  solar radius = 1.00 Jupiter radius.
- 50. Answer: 38.** The slope of recorded counts versus time is  $(2120-620)/(40-10) = 50$  counts/s. Fixed bias affects the intercept, not the slope. Subtract dark-current rate: source rate =  $50-12 = 38$  counts/s.

## Mock Paper 2 - Answer Key and Solutions

### Section A Answer Key

| 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. B  | 2. D  | 3. A  | 4. C  | 5. B  | 6. D  | 7. A  | 8. C  | 9. B  | 10. D |
| 11. A | 12. C | 13. B | 14. D | 15. A | 16. C | 17. B | 18. D | 19. A | 20. C |
| 21. B | 22. D | 23. A | 24. C | 25. B | 26. D | 27. A | 28. C | 29. B | 30. D |
| 31. A | 32. C | 33. B | 34. D | 35. A | 36. C | 37. B | 38. D | 39. A | 40. C |

### Section B Numerical Answers

| Q41-Q42               | Q43-Q44               | Q45-Q46             | Q47-Q48               | Q49-Q50             |
|-----------------------|-----------------------|---------------------|-----------------------|---------------------|
| 41: 42200<br>42: 12.5 | 43: 0.315<br>44: 5.54 | 45: 259<br>46: 0.99 | 47: 1.64<br>48: 180.0 | 49: 18.28<br>50: 52 |

### Concise Solutions - Section A

- Answer: B.** A spectrograph disperses light by wavelength. Line positions identify composition and Doppler shifts give radial motion.
- Answer: D.** The L1 region offers a nearly uninterrupted solar view and upstream measurements of solar particles and fields.
- Answer: A.** Passive remote sensing detects naturally available radiation, such as reflected solar light.
- Answer: C.** Hydrogen and oxygen are gases at room temperature; liquefaction permits compact storage and high-performance propulsion.
- Answer: B.** Flat rotation curves indicate more gravitating mass than is accounted for by visible stars and gas.
- Answer: D.**  $d(\text{pc})=1/p=20$  pc.
- Answer: A.** Hour angle  $H=LST-RA=5$  h. Positive H means west of the meridian.
- Answer: C.**  $h_{\text{max}}=90$  degrees- $|\text{latitude-declination}|=50$  degrees.
- Answer: B.**  $\theta=1.22 \lambda/D=0.503$  arcsecond.
- Answer: D.**  $v/c=\Delta \lambda/\lambda$ . This gives  $v=240$  km/s, away from us.
- Answer: A.**  $F_A/F_B=10^{(0.4 \Delta m)}=3.98$ .
- Answer: C.**  $N=A F t \eta/(hc/\lambda)=1.932e+08$ .
- Answer: B.**  $R=\lambda/\Delta \lambda=5000$ .
- Answer: D.** Collected photons scale as  $D^2 t$ , so  $t_2=t_1(D_1/D_2)^2=40$  s.
- Answer: A.** In solar units,  $P^2=a^3$ , hence  $P=3.38$  years.
- Answer: C.**  $E=-GMm/(2r)$ .  $\Delta E=(GMm/2r)(1-1/3)=0.333 GMm/r$ .
- Answer: B.**  $e=(Q-q)/(Q+q)=0.733$ .
- Answer: D.**  $1/S=|1/P_1-1/P_2|$ , so  $S=2.14$  years.
- Answer: A.**  $H=kT/(\mu g)$ , so  $H_B/H_A=1.5/(3 \times 0.5)=1$ .
- Answer: C.**  $\rho$  proportional to  $M/R^3$ , so ratio  $=10/2^3=1.25$ .
- Answer: B.**  $\delta=(R_p/R_s)^2$ , so  $R_p/R_s=\sqrt{0.0081}=0.09$ .
- Answer: D.**  $T_{\text{eq}}$  approximately  $278 K L^{(1/4)}/\sqrt{a}=278 K$ .
- Answer: A.**  $L/L_{\text{sun}}=(R/R_{\text{sun}})^2(T/T_{\text{sun}})^4=4.744$ .
- Answer: C.**  $\tau$  proportional to  $M/L=M^{(1-3.5)}$ , so  $\tau=10 \times 3^{(1-3.5)}=0.64$  billion years.
- Answer: B.** In solar units,  $M_{\text{total}}=a^3/P^2=4 M_{\text{sun}}$ .
- Answer: D.**  $R_s$  approximately  $3(M/M_{\text{sun}}) \text{ km}=24 \text{ km}$ .
- Answer: A.**  $z=(\lambda_{\text{obs}}-\lambda_{\text{emit}})/\lambda_{\text{emit}}=0.15$ .
- Answer: C.** From  $v^2=GM(r)/r$ , constant  $v$  requires  $M(r)$  proportional to  $r$ .
- Answer: B.** Tidal gradient scales as  $1/r^3$ , giving  $1/2.5^3=0.064$ .
- Answer: D.**  $R \text{ ratio}=64^{(-1/6)}=0.5$ .

31. **Answer: A.** Round-trip time= $2d/c=8$  s.
32. **Answer: C.** GSD approximately  $H/p/f=2.5$  m per pixel.
33. **Answer: B.**  $P=A S \eta \cos(\theta)=4240.1$  W.
34. **Answer: D.**  $\tau=\Delta L/\Delta t=I \Delta \omega/t=0.9$  N m.
35. **Answer: A.**  $m_0/m_f=\exp(\Delta v/(I_{sp} g_0))=4.62$ .
36. **Answer: C.**  $F=\dot{m} v_e+(p_e-p_a)A_e=795000$  N.
37. **Answer: B.** SpaDeX demonstrated rendezvous, docking, undocking and later power transfer between two spacecraft.
38. **Answer: D.** Passivation vents residual propellants and discharges batteries, reducing fragmentation risk.
39. **Answer: A.** Europa Clipper investigates the ice shell, subsurface ocean, composition and geology to assess habitability.
40. **Answer: C.** Photon energy= $1240/\lambda=2.76$  eV;  $K_{\max}=E-\phi=0.66$  eV.

## Worked Solutions - Section B

41. **Answer: 42200.**  $r=[\mu T^2/(4\pi^2)]^{1/3}=42164$  km, which rounds to 42200 km.
42. **Answer: 12.5.**  $d=1/p=12.5$  pc.
43. **Answer: 0.315.**  $D=1.22 \lambda/\theta=0.315$  m.
44. **Answer: 5.54.**  $m_0/m_f=\exp(\Delta v/(I_{sp} g_0))=5.54$ .
45. **Answer: 259.** Transfer is half an ellipse of semi-major axis 1.262 AU:  $t=0.5 a^{3/2}=258.9$  days.
46. **Answer: 0.99.**  $R_p/R_s=\sqrt{0.0121}$ ;  $R_p=0.99 R_j$ .
47. **Answer: 1.64.**  $L/L_{\text{sun}}=R^2 T^4=1.64$ .
48. **Answer: 180.0.**  $d=v/H_0=180.0$  Mpc.
49. **Answer: 18.28.**  $P=AS\eta=3046.4$  W;  $E=Pt=18.28$  kWh.
50. **Answer: 52.** Slope=60 counts/s; bias affects intercept. Source rate=slope-dark=52 counts/s.

# Mock Paper 3 - Answer Key and Solutions

## Section A Answer Key

| 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C  | 2. A  | 3. D  | 4. B  | 5. C  | 6. A  | 7. D  | 8. B  | 9. C  | 10. A |
| 11. D | 12. B | 13. C | 14. A | 15. D | 16. B | 17. C | 18. A | 19. D | 20. B |
| 21. C | 22. A | 23. D | 24. B | 25. C | 26. A | 27. D | 28. B | 29. C | 30. A |
| 31. D | 32. B | 33. C | 34. A | 35. D | 36. B | 37. C | 38. A | 39. D | 40. B |

## Section B Numerical Answers

| Q41-Q42              | Q43-Q44              | Q45-Q46              | Q47-Q48              | Q49-Q50              |
|----------------------|----------------------|----------------------|----------------------|----------------------|
| 41: 2.38<br>42: 97.1 | 43: 300.0<br>44: 400 | 45: 7.3<br>46: 0.240 | 47: 0.02<br>48: 12.5 | 49: 4.00<br>50: 1.99 |

## Concise Solutions - Section A

- Answer: C.** Photometry measures received flux or brightness, usually through one or more filters.
- Answer: A.** Accelerating a wheel produces an equal and opposite change in spacecraft angular momentum.
- Answer: D.** The habitable-zone concept concerns conditions compatible with surface liquid water, not a guarantee of life.
- Answer: B.** The L1 geometry supports sustained solar observation and upstream measurements relevant to space weather.
- Answer: C.** Reusability can reduce hardware replacement costs, although refurbishment and operations still matter.
- Answer: A.**  $d(\text{pc})=1/p=5$  pc.
- Answer: D.**  $\theta=D/d=9.115\text{e-}03$  rad=0.522 degree.
- Answer: B.**  $\theta=1.22 \lambda/D=0.755$  arcsecond.
- Answer: C.**  $v/c=\Delta \lambda/\lambda$ . This gives  $v=182.8$  km/s, towards us.
- Answer: A.**  $F_A/F_B=10^{(0.4 \Delta m)}=100$ .
- Answer: D.**  $N=A F t \eta/(hc/\lambda)=5.795\text{e}+08$ .
- Answer: B.**  $R=\lambda/\Delta \lambda=10000$ .
- Answer: C.**  $h_{\text{max}}=90 \text{ degrees}-|\text{latitude-declination}|=60$  degrees.
- Answer: A.** In solar units,  $P^2=a^3$ , hence  $P=15.62$  years.
- Answer: D.**  $v_{\text{esc}}$  proportional to  $r^{(-1/2)}$ , giving 0.707.
- Answer: B.**  $E=-GMm/(2r)$ .  $\Delta E=(GMm/2r)(1-1/4)=0.375 GMm/r$ .
- Answer: C.**  $e=(Q-q)/(Q+q)=0.6$ .
- Answer: A.**  $1/S=|1/P_1-1/P_2|$ , so  $S=1.6$  years.
- Answer: D.**  $H=kT/(\mu g)$ , so  $H_B/H_A=2/(4 \times 1)=0.5$ .
- Answer: B.**  $\delta=(R_p/R_s)^2$ , so  $R_p/R_s=\sqrt{0.0225}=0.15$ .
- Answer: C.**  $T_{\text{eq}}$  approximately  $278 \text{ K } L^{(1/4)}/\sqrt{a}=278 \text{ K}$ .
- Answer: A.**  $L/L_{\text{sun}}=(R/R_{\text{sun}})^2(T/T_{\text{sun}})^4=4$ .
- Answer: D.**  $\tau$  proportional to  $M/L=M^{(1-4)}$ , so  $\tau=10 \times 5^{(1-4)}=0.08$  billion years.
- Answer: B.** In solar units,  $M_{\text{total}}=a^3/P^2=12 M_{\text{sun}}$ .
- Answer: C.**  $R_s$  approximately  $3(M/M_{\text{sun}}) \text{ km}=60 \text{ km}$ .
- Answer: A.**  $v=H_0 d=17000 \text{ km/s}$ .
- Answer: D.**  $z=(\lambda_{\text{obs}}-\lambda_{\text{emit}})/\lambda_{\text{emit}}=3$ .
- Answer: B.** Tidal gradient scales as  $1/r^3$ , giving  $1/3^3=0.037$ .
- Answer: C.**  $v_{\text{esc}}$  proportional to  $\sqrt{gR}$ , giving  $\sqrt{0.16 \times 0.27}=0.208$ .
- Answer: A.**  $R \text{ ratio}=729^{(-1/6)}=0.333$ .

31. **Answer: D.**  $\Pr$  proportional to  $Dt^2 Dr^2/R^2$ , giving 0.141.
32. **Answer: B.**  $\text{Storage} = \text{rate} \times \text{time} / 8 = 16.2 \text{ GB}$ .
33. **Answer: C.**  $\tau = \Delta L / \Delta t = I \Delta \omega / t = 0.5 \text{ N m}$ .
34. **Answer: A.**  $I_{sp} = F / (\dot{m} g_0) = 306.1 \text{ s}$ .
35. **Answer: D.** Mass ratios multiply:  $4 \times 3.5 = 14$ .
36. **Answer: B.** A gravity turn lets the flight path rotate toward the horizontal with relatively small steering losses.
37. **Answer: C.** The air-drop test validates the parachute sequence and recovery of a crew-module analogue.
38. **Answer: A.** The mission uses many flybys to investigate the moon's ice, composition, geology and habitability-related ocean evidence.
39. **Answer: D.** A controlled experiment changes the independent variable while holding other relevant variables fixed.
40. **Answer: B.**  $|M| = f_o / f_e = 30$ .

## Worked Solutions - Section B

41. **Answer: 2.38.**  $v_{esc} = \sqrt{2\mu/R} = 2.38 \text{ km/s}$ .
42. **Answer: 97.1.**  $T = 2\pi \sqrt{r^3/\mu} = 97.1 \text{ min}$ .
43. **Answer: 300.0.**  $v = c \Delta \lambda / \lambda = 300.0 \text{ km/s}$ .
44. **Answer: 400.**  $\lambda_{max} = b/T = 399.7 \text{ nm}$ .
45. **Answer: 7.3.**  $H = kT / (\mu m_u g) = 7.3 \text{ km}$ .
46. **Answer: 0.240.**  $\tau = I \Delta \omega / t = 0.240 \text{ N m}$ .
47. **Answer: 0.02.**  $p_{eff} = \beta m v = 1.584 \times 10^7 \text{ kg m/s} = 0.02 \times 10^9$ .
48. **Answer: 12.5.**  $t = d/c = 12.5 \text{ min}$ .
49. **Answer: 4.00.**  $E = hc/\lambda = 1240/310 = 4.00 \text{ eV}$ .
50. **Answer: 1.99.**  $b = \sum[(x - \bar{x})(y - \bar{y})] / \sum[(x - \bar{x})^2] = 1.99$ .

# Mock Paper 4 - Answer Key and Solutions

## Section A Answer Key

| 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. D  | 2. C  | 3. B  | 4. A  | 5. D  | 6. C  | 7. B  | 8. A  | 9. D  | 10. C |
| 11. B | 12. A | 13. D | 14. C | 15. B | 16. A | 17. D | 18. C | 19. B | 20. A |
| 21. D | 22. C | 23. B | 24. A | 25. D | 26. C | 27. B | 28. A | 29. D | 30. C |
| 31. B | 32. A | 33. D | 34. C | 35. B | 36. A | 37. D | 38. C | 39. B | 40. A |

## Section B Numerical Answers

| Q41-Q42              | Q43-Q44              | Q45-Q46                | Q47-Q48               | Q49-Q50             |
|----------------------|----------------------|------------------------|-----------------------|---------------------|
| 41: 35800<br>42: 473 | 43: 0.98<br>44: 3.24 | 45: 250.0<br>46: 30.00 | 47: 6.42<br>48: 59.53 | 49: 161.2<br>50: 50 |

## Concise Solutions - Section A

- Answer: D.** Radar is active remote sensing because the instrument supplies the illuminating signal and measures the return.
- Answer: C.** Lensing measures the total gravitating mass along the line of sight, including matter that does not emit detectable light.
- Answer: B.** High combustion temperature and low molecular mass of the exhaust support a high effective exhaust velocity.
- Answer: A.** The spacecraft is deflected in the planet frame; because the planet itself moves around the Sun, heliocentric energy can be exchanged.
- Answer: D.** A collision cascade can increase debris density and raise the probability of additional impacts.
- Answer: C.**  $m-M=5 \log_{10}(d/10 \text{ pc})$ , giving  $d=1000 \text{ pc}$ .
- Answer: B.** Hour angle  $H=LST-RA=8 \text{ h}$ . Positive  $H$  means west of the meridian.
- Answer: A.**  $h_{\text{max}}=90 \text{ degrees}-|\text{latitude-declination}|=60 \text{ degrees}$ .
- Answer: D.**  $\theta=D/d=9.291\text{e-}03 \text{ rad}=0.532 \text{ degree}$ .
- Answer: C.**  $\theta=1.22 \lambda/D=0.277 \text{ arcsecond}$ .
- Answer: B.**  $v/c=\Delta \lambda/\lambda$ . This gives  $v=898.6 \text{ km/s}$ , away from us.
- Answer: A.**  $F_A/F_B=10^{(0.4 \Delta m)}=2$ .
- Answer: D.**  $R=\lambda/\Delta \lambda=32815$ .
- Answer: C.** In solar units,  $P^2=a^3$ , hence  $P=0.12 \text{ years}$ .
- Answer: B.**  $v_{\text{esc}}$  proportional to  $r^{(-1/2)}$ , giving 0.447.
- Answer: A.**  $E=-GMm/(2r)$ .  $\Delta E=(GMm/2r)(1-1/2.5)=0.3 GMm/r$ .
- Answer: D.**  $e=(Q-q)/(Q+q)=0.88$ .
- Answer: C.**  $1/S=|1/P_1-1/P_2|$ , so  $S=1.09 \text{ years}$ .
- Answer: B.**  $H=kT/(\mu g)$ , so  $H_B/H_A=0.8/(0.4 \times 2)=1$ .
- Answer: A.**  $\delta=(R_p/R_s)^2$ , so  $R_p/R_s=\sqrt{0.0036}=0.06$ .
- Answer: D.**  $T_{\text{eq}}$  approximately  $278 \text{ K } L^{(1/4)}/\sqrt{a}=278 \text{ K}$ .
- Answer: C.**  $L/L_{\text{sun}}=(R/R_{\text{sun}})^2(T/T_{\text{sun}})^4=2.161$ .
- Answer: B.**  $\tau$  proportional to  $M/L=M^{(1-3)}$ , so  $\tau=10 \times 2.5^{(1-3)}=1.6 \text{ billion years}$ .
- Answer: A.** In solar units,  $M_{\text{total}}=a^3/P^2=5 M_{\text{sun}}$ .
- Answer: D.**  $R_s$  approximately  $3(M/M_{\text{sun}}) \text{ km}=3\text{e}+06 \text{ km}$ .
- Answer: C.**  $v=H_0 d=5760 \text{ km/s}$ .
- Answer: B.**  $z=(\lambda_{\text{obs}}-\lambda_{\text{emit}})/\lambda_{\text{emit}}=0.5$ .
- Answer: A.** The supernova distance-redshift relation revealed accelerated expansion, commonly modelled with dark energy.
- Answer: D.** Tidal gradient scales as  $1/r^3$ , giving  $1/1.5^3=0.296$ .

30. **Answer: C.**  $R \text{ ratio} = 4^{(-1/6)} = 0.794$ .

31. **Answer: B.** Round-trip time  $= 2d/c = 2.56 \text{ s}$ .

32. **Answer: A.** GSD approximately  $H p/f = 6.25 \text{ m}$  per pixel.

33. **Answer: D.** Storage  $= \text{rate} \times \text{time} / 8 = 32.4 \text{ GB}$ .

34. **Answer: C.**  $P_r$  proportional to  $Dt^2 D_r^2 / R^2$ , giving 1.266.

35. **Answer: B.**  $m_0/m_f = \exp(\Delta v / (I_{sp} g_0)) = 5.48$ .

36. **Answer: A.**  $F = m \dot{v} + (p_e - p_a) A_e = 751000 \text{ N}$ .

37. **Answer: D.**  $p = mv = 500 \times 12000 = 6.000 \times 10^6 \text{ kg m/s}$ .

38. **Answer: C.** Photon energy  $= 1240/\lambda = 3.1 \text{ eV}$ ;  $K_{\max} = E - \phi = 1.1 \text{ eV}$ .

39. **Answer: B.** For maximum fractional uncertainty in a product, percentages add:  $3 + 4 = 7\%$ .

40. **Answer: A.** Aditya-L1 combines remote observations of the solar atmosphere with local measurements of particles and magnetic fields.

## Worked Solutions - Section B

41. **Answer: 35800.** Orbital radius is 42164 km, so altitude is 35786 km, or 35800 km to the nearest 100 km.

42. **Answer: 473.** Transfer is half an ellipse of semi-major axis 1.885 AU:  $t = 0.5 a^{3/2} = 472.6 \text{ days}$ .

43. **Answer: 0.98.**  $R_p/R_s = \sqrt{0.0049}$ ;  $R_p = 0.98 R_j$ .

44. **Answer: 3.24.**  $L/L_{\text{sun}} = R^2 T^4 = 3.24$ .

45. **Answer: 250.0.**  $d = v/H_0 = 250.0 \text{ Mpc}$ .

46. **Answer: 30.00.**  $P = A S_{\text{eta}} = 3750.0 \text{ W}$ ;  $E = Pt = 30.00 \text{ kWh}$ .

47. **Answer: 6.42.**  $m_0/m_f = \exp(\Delta v / (I_{sp} g_0)) = 6.42$ .

48. **Answer: 59.53.**  $v_{\text{esc}} = \sqrt{2\mu/R} = 59.53 \text{ km/s}$ .

49. **Answer: 161.2.**  $H = kT / (\mu m_u g) = 161.2 \text{ km}$ .

50. **Answer: 50.** Slope  $= 55 \text{ counts/s}$ ; bias affects intercept. Source rate  $= \text{slope} - \text{dark} = 50 \text{ counts/s}$ .

# Mock Paper 5 - Answer Key and Solutions

## Section A Answer Key

| 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. A  | 2. C  | 3. B  | 4. D  | 5. A  | 6. C  | 7. B  | 8. D  | 9. A  | 10. C |
| 11. B | 12. D | 13. A | 14. C | 15. B | 16. D | 17. A | 18. C | 19. B | 20. D |
| 21. A | 22. C | 23. B | 24. D | 25. A | 26. C | 27. B | 28. D | 29. A | 30. C |
| 31. B | 32. D | 33. A | 34. C | 35. B | 36. D | 37. A | 38. C | 39. B | 40. D |

## Section B Numerical Answers

| Q41-Q42             | Q43-Q44              | Q45-Q46               | Q47-Q48              | Q49-Q50              |
|---------------------|----------------------|-----------------------|----------------------|----------------------|
| 41: 40<br>42: 0.818 | 43: 500<br>44: 235.2 | 45: 6.04<br>46: 1.000 | 47: 0.03<br>48: 18.3 | 49: 2.00<br>50: 2.05 |

## Concise Solutions - Section A

- Answer: A.** A spectrograph forms a wavelength-resolved record from which line positions and strengths can be measured.
- Answer: C.** The habitable zone is only an orbital-energy criterion; atmosphere, albedo, pressure and other environmental factors matter.
- Answer: B.** Conservation of angular momentum gives the body an equal and opposite angular-momentum change.
- Answer: D.** The experiment concerned autonomous docking and transfer of electrical power between the two spacecraft.
- Answer: A.** Recovery is useful only if inspection, refurbishment and operations permit reliable, sufficiently frequent reuse.
- Answer: C.**  $d(\text{pc})=1/p=25 \text{ pc}$ .
- Answer: B.** Hour angle  $H=LST-RA=21.25 \text{ h}$ . Positive  $H$  means west of the meridian.
- Answer: D.**  $h_{\text{max}}=90 \text{ degrees}-|\text{latitude-declination}|=40 \text{ degrees}$ .
- Answer: A.**  $\theta=1.22 \lambda/D=0.126 \text{ arcsecond}$ .
- Answer: C.**  $v/c=\Delta \lambda/\lambda$ . This gives  $v=925.7 \text{ km/s}$ , away from us.
- Answer: B.**  $F_A/F_B=10^{(0.4 \Delta m)}=19.05$ .
- Answer: D.**  $N=A F t \eta/(hc/\lambda)=8.452 \times 10^7$ .
- Answer: A.**  $R=\lambda/\Delta \lambda=12160$ .
- Answer: C.** In solar units,  $P^2=a^3$ , hence  $P=31.62 \text{ years}$ .
- Answer: B.**  $v_{\text{esc}}$  proportional to  $r^{(-1/2)}$ , giving 0.833.
- Answer: D.**  $E=-GMm/(2r)$ .  $\Delta E=(GMm/2r)(1-1/10)=0.45 GMm/r$ .
- Answer: A.**  $e=(Q-q)/(Q+q)=0.96$ .
- Answer: C.**  $1/S=|1/P_1-1/P_2|$ , so  $S=0.4 \text{ years}$ .
- Answer: B.**  $H=kT/(\mu g)$ , so  $H_B/H_A=3/(6 \times 0.25)=2$ .
- Answer: D.**  $\rho$  proportional to  $M/R^3$ , so ratio  $=20/2.2^3=1.878$ .
- Answer: A.**  $\delta=(R_p/R_s)^2$ , so  $R_p/R_s=\sqrt{0.0004}=0.02$ .
- Answer: C.**  $T_{\text{eq}}$  approximately  $278 \text{ K } L^{(1/4)}/\sqrt{a}=197 \text{ K}$ .
- Answer: B.**  $L/L_{\text{sun}}=(R/R_{\text{sun}})^2(T/T_{\text{sun}})^4=24.01$ .
- Answer: D.**  $\tau$  proportional to  $M/L=M^{(1-3.5)}$ , so  $\tau=10 \times 8^{(1-3.5)}=0.06 \text{ billion years}$ .
- Answer: A.** In solar units,  $M_{\text{total}}=a^3/P^2=24 M_{\text{sun}}$ .
- Answer: C.**  $R_s$  approximately  $3(M/M_{\text{sun}}) \text{ km}=1.2 \times 10^7 \text{ km}$ .
- Answer: B.**  $v=H_0 d=33700 \text{ km/s}$ .
- Answer: D.**  $z=(\lambda_{\text{obs}}-\lambda_{\text{emit}})/\lambda_{\text{emit}}=1.5$ .
- Answer: A.** Tidal gradient scales as  $1/r^3$ , giving  $1/4^3=0.016$ .
- Answer: C.**  $v_{\text{esc}}$  proportional to  $\sqrt{gR}$ , giving  $\sqrt{0.38 \times 0.53}=0.449$ .

31. **Answer: B.**  $R \text{ ratio} = 15625^{(-1/6)} = 0.2$ .
32. **Answer: D.** GSD approximately  $H \text{ p/f} = 1.4 \text{ m per pixel}$ .
33. **Answer: A.**  $P = A S \eta \cos(\theta) = 4352 \text{ W}$ .
34. **Answer: C.**  $\text{Storage} = \text{rate} \times \text{time} / 8 = 33.75 \text{ GB}$ .
35. **Answer: B.**  $\tau = \Delta L / \Delta t = I \Delta \omega / t = 4 \text{ N m}$ .
36. **Answer: D.**  $I_{sp} = F / (\dot{m} g_0) = 306.1 \text{ s}$ .
37. **Answer: A.** Mass ratios multiply:  $5 \times 4 = 20$ .
38. **Answer: C.** Aditya-L1 observes the solar atmosphere and measures particles and magnetic fields near L1.
39. **Answer: B.** A subsurface ocean and chemical-energy possibilities make Europa an important habitability target.
40. **Answer: D.**  $|M| = f_o / f_e = 40$ .

## Worked Solutions - Section B

41. **Answer: 40.**  $d = 1/p = 40 \text{ pc}$ .
42. **Answer: 0.818.**  $D = 1.22 \lambda / \theta = 0.818 \text{ m}$ .
43. **Answer: 500.**  $\lambda_{\text{max}} = b/T = 499.7 \text{ nm}$ .
44. **Answer: 235.2.**  $T = 2\pi \sqrt{r^3/\mu} = 235.2 \text{ min}$ .
45. **Answer: 6.04.**  $m_0/m_f = \exp(\Delta v / (I_{sp} g_0)) = 6.04$ .
46. **Answer: 1.000.**  $\tau = I \Delta \omega / t = 1.000 \text{ N m}$ .
47. **Answer: 0.03.**  $p_{\text{eff}} = \beta m v = 3.150 \times 10^7 \text{ kg m/s} = 0.03 \times 10^9$ .
48. **Answer: 18.3.**  $t = d/c = 18.3 \text{ min}$ .
49. **Answer: 2.00.**  $E = hc/\lambda = 1240/620 = 2.00 \text{ eV}$ .
50. **Answer: 2.05.**  $b = \sum[(x - \bar{x})(y - \bar{y})] / \sum[(x - \bar{x})^2] = 2.05$ .

## Syllabus Coverage Audit

The table below counts the questions assigned to each syllabus unit in each paper. Integrated problems are assigned to the unit that supplies the principal assessed concept. Every paper samples all major strands, and the five-paper set covers all eleven supplied units repeatedly.

| Unit | Syllabus area                           | P1 | P2 | P3 | P4 | P5 |
|------|-----------------------------------------|----|----|----|----|----|
| 1    | Fundamentals of Astronomy               | 13 | 12 | 11 | 8  | 12 |
| 2    | The Solar System                        | 14 | 10 | 10 | 12 | 9  |
| 3    | Stars, Galaxies & the Universe          | 11 | 9  | 6  | 10 | 6  |
| 4    | Earth & Moon                            | 1  | 2  | 4  | 2  | 4  |
| 5    | Satellites & Space Technology           | 3  | 7  | 6  | 7  | 7  |
| 6    | Rockets & Launch Vehicles               | 2  | 4  | 3  | 4  | 3  |
| 7    | ISRO & India's Space Missions           | 2  | 2  | 2  | 1  | 2  |
| 8    | Space Exploration                       | 0  | 1  | 3  | 2  | 3  |
| 9    | Basic Physics for Space Science         | 2  | 1  | 2  | 1  | 2  |
| 10   | Scientific Reasoning & Logical Aptitude | 1  | 1  | 2  | 2  | 1  |
| 11   | General Space Awareness                 | 1  | 1  | 1  | 1  | 1  |

### Difficulty Design

Each paper contains only five direct warm-up questions. Questions 6-40 are predominantly multi-step or application-based single-correct questions at the intended JEE Main benchmark. Questions 41-50 require numerical computation or quantitative data analysis.

### Current-information verification note

Mission-status statements used in this volume were checked against official ISRO and NASA mission pages. SpaDeX is treated as a demonstrated rendezvous, docking, undocking and power-transfer technology mission; Aditya-L1 as an operational solar observatory near Sun-Earth L1 with released science data; Gaganyaan as a programme under development and testing, including parachute and recovery validation; and Europa Clipper as an active mission en route to Jupiter for investigation of Europa's habitability. No planned mission is described as completed.

### End of volume.